Internet of Things 38 (2026) 101996



Contents lists available at ScienceDirect

Internet of Things

journal homepage: www.elsevier.com/locate/iot





FedMP-LDL: Lightweight DDoS detection and mitigation in SDN-Fog IoT networks

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ARTICLE INFO

Keywords:
Attack detection
Mitigation
Fuzzy-rule
Controller
Federated learning
Privacy-preserving

ABSTRACT

Software-Defined Networking (SDN) integrated with fog-enabled internet of things environments improves scalability, flexibility, and centralized network management. However, the presence of heterogeneous IoT devices and resource-constrained fog nodes significantly increases vulnerability to large-scale Distributed Denial-of-Service (DDoS) attacks. Existing DDoS detection solutions often rely on centralized learning or computationally intensive deep learning models, which leads to higher latency and reduced real-time applicability. To address these issues, this research proposes a Federated Multi-Phase Lightweight Deep Learning (FedMP-LDL) framework for efficient DDoS detection and mitigation in SDN–fog–IoT networks. The novelty of FedMP-LDL is derived from its multi-phase architecture integrating Hamacher-based fuzzy equivalence rules for lightweight anomaly detection and a progressive distilled ghost-squeeze excitation network for hierarchical federated learning across fog nodes. In addition, multi-objective reinforcement learning is incorporated to enable adaptive SDN flow-rule optimization and dynamic DDoS mitigation in fog-enabled IoT environments. This distributed strategy eliminates dependency on centralized training, enhances scalability, and maintains robust performance under varying attack intensities. Experimental results demonstrate that FedMP-LDL achieves 99.72% accuracy, 99.66% precision, 99.57% recall, and 99.61% F1-score with a low inference time of 0.1762 s, outperforming existing state-of-the-art approaches while ensuring an effective balance between detection accuracy and computational efficiency.

1. Introduction

Various physical objects are integrated into a digital environment to enable seamless internet-based connectivity and interaction based on Internet of Things (IoT) [1]. IoT become the basis for cutting-edge applications across numerous industries, with billions of smart devices are used in worldwide. By facilitating real-time data collection, wide connectivity, and intelligent decision-making, the IoT is transforming various industrial sectors, including smart cities, agriculture, healthcare, industrial automation, supply chain management, intelligent transportation systems, smart grids, autonomous surveillance, e-commerce infrastructures, and smart home automation. These advancements improve operational efficiency, resource management, and user convenience, making IoT a crucial component of modern digital infrastructure. In particular, SDN-fog-enabled IoT architectures support low-latency communication,

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https://doi.org/10.1016/j.iot.2026.101996
Received 8 December 2025; Received in revised form 27 May 2026; Accepted 4 June 2026
Available online 8 June 2026
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dynamic resource allocation, and continuous service availability for large-scale distributed applications. However, the rapid growth of heterogeneous IoT devices and distributed fog nodes introduces complex networking environments where security, scalability, and interoperability become critical challenges. Furthermore, the increasing connectivity of resource-constrained devices significantly increases vulnerability to cyberattacks, especially DDoS attacks, which overwhelm SDN controllers and fog nodes, severely disrupting critical services and compromising overall network reliability.

IoT is inherently vulnerable, because of its unique characteristics and transformative potential. Advanced security measures are challenging to implement on the majority of IoT devices due to their limited computational power memory and energy [2]. Furthermore, the creation of common security standards is hampered by the variety of devices and communication protocols. Manufacturers often introduce systemic flaws by putting market competitiveness and quick deployment ahead of thorough security testing. Due to these vulnerabilities IoT infrastructures are susceptible to a variety of cyberthreats such as large-scale DDoS attacks data breaches and illegal access. These attacks endanger the availability of services hinder essential operations and cause cascading failures across linked networks. By separating the control plane from the data plane and enabling centralized programmable and flexible network control SDN offers a disruptive approach to contemporary network administration [3]. Because, it enables better network monitoring, dynamic resource allocation and increased scalability, this architectural paradigm is perfect for Internet of Things applications. The application control and data layers make up the SDN framework. While the data layer controls packet forwarding via SDN-enabled switches the application layer establishes network policies and services that are dynamically enforced by an SDN controller in the control layer. Data from IoT devices is routed through switches and gateways in SDN-enabled IoT systems and the controller controls network activity in real time [4].

However, SDN controllers are centralized, which creates additional security risks, since attackers could use control-plane vulnerabilities to launch massive DDoS attacks and overload network infrastructure. Recent research has moved toward Machine Learning (ML), Deep Learning (DL) and Federated Learning (FL) approaches to improve detection accuracy and resilience to evolving threats in order to overcome these challenges [5]. A reinforcement learning-based framework is developed using dilated convolution layers for feature selection, class balancing, and improved model generalization. In addition, Shapley additive explanations (SHAP) guided feature identification, Bayesian Optimization Hyper Band (BOHB) based hyperparameter optimization and Generative Adversarial Network (GAN) based dynamic data augmentation are employed to increase robustness and interpretability [6]. However, the framework suffers from high computational complexity, increased training overhead, and reduced real-time applicability in large-scale SDN-IoT environments. To improve training stability and detection performance a Scope Loss Function (SLF) is also used to balance the phases of exploration and exploitation. Similarly, to increase diagnostic reliability and classification accuracy, EfficientNetB3-based Convolutional Neural Network (CNN) uses a confidence-driven hybrid decision mechanism that preserves high-confidence predictions and selectively substitutes expert human evaluations for low-confidence outputs [7]. Nevertheless, these approaches require high computational resources, depend heavily on confidence threshold selection, and are difficult to deploy in resource-constrained fog-IoT systems.

AI-driven, FL, and fog-enabled security frameworks have been widely investigated for DDoS detection and intrusion mitigation in SDN-IoT environments. Secure 5G-SDN frameworks integrating federated reinforcement learning have been developed for anomaly detection, mitigation, and attack forensics [8]. However, frequent federated updates and reinforcement learning optimization increase communication overhead and training complexity. Federated edge intelligence-based frameworks have also been introduced for real-time cyberattack detection in heterogeneous IoT systems [9]. These methods still suffer from scalability limitations and inconsistent performance under highly dynamic IoT traffic conditions. Advanced AI-driven security risk detection frameworks have been proposed for heterogeneous cloud-fog computing networks to improve intelligent threat analysis and distributed security management [10]. However, the integration of multiple AI modules increases model complexity and resource consumption. Federated continual learning-based intrusion detection models have been designed for secure fog computing environments to combat emerging cyberattacks using benchmark datasets [11]. Nevertheless, continual retraining increases computational cost and lead to model drift under evolving attack patterns. In addition, multilayer reputation-based defense mechanisms have been proposed for FL in edge-fog-cloud IoT systems to strengthen distributed trust management and attack resilience [12]. However, reputation management mechanisms introduce additional communication latency and reduce real-time mitigation efficiency in large-scale IoT environments. Although these approaches improve distributed learning, anomaly detection, and security management, several challenges still remain, including high computational complexity, communication overhead, scalability limitations, adaptive real-time mitigation, and efficient deployment in resource-constrained SDN-fog IoT environments. To address these limitations, the proposed FedMP-LDL framework introduces a lightweight multi-phase architecture integrating Hamacher FERB-based anomaly filtering and progressively distilled Ghost-SE federated learning. In addition, reinforcement learning-driven dynamic SDN flow adaptation is incorporated to achieve scalable, privacy-preserving, low-latency, and computationally efficient DDoS detection and mitigation. "The proposed FedMP-LDL framework enables lightweight, scalable, privacy-preserving, and real-time DDoS detection and mitigation in SDN-fog-enabled IoT environments with reduced computational complexity and enhanced adaptive learning capability".

1.1. Motivation and contribution

Despite the effectiveness of SDN-fog designs in managing consumer IoT traffic, they bring additional risks. The centralized controller becomes a prime target for assaults, and the large number of resource-constrained IoT devices exacerbates threats such as monitoring, exploitation, and availability attacks [13]. DDoS assaults are particularly dangerous because they use botnets of compromised devices to flood network resources with illicit traffic, overwhelming SDN controllers and fog nodes equally. This causes severe service disruption in key areas such as smart cities, healthcare, and commercial infrastructures. Effective detection and mitigation of

DDoS assaults in SDN-based IoT environments remains a critical and hard field of research. The primary contributions and innovation of this research are summarized as follows:

- This research proposed the novel FedMP-LDL framework, a unified multi-phase workflow integrating Hamacher FERB based anomaly filtering, privacy-preserving federated classification, and adaptive SDN mitigation to secure SDN-fog IoT networks against large-scale DDoS attacks
- For lightweight anomaly detection an innovative Fuzzy Equivalence Rule-Based (FERB) model based on Hamacher equivalence and Kernel Principal Component Analysis-Temporal Convolutional Network Transformer (KPCA-TCN) is used to capture uncertain traffic patterns and temporal dependencies at fog nodes with limited resources
- Progressively distilled GhostNet-Squeeze Excitation (Ghost-SE) model is designed with Animated Oat Optimization (AOO) and aggregated through Hierarchical Federated Averaging (H-FedAvg), achieving 67% parameter reduction with greater than 99% accuracy
- The novel Reinforcement Learning–Driven Dynamic Flow Rule Adaptation (RL-DFRA) mechanism is implemented, featuring multi-objective Q-learning at the SDN controller for continuous, context-aware dynamic flow rule adaptation ensuring scalable DDoS mitigation

Overall, the proposed FedMP-LDL framework combines lightweight detection, efficient federated learning, and adaptive RL-based mitigation to achieve scalable, high-accuracy DDoS defense with reduced computational complexity in SDN-fog IoT environments.
Organization: The rest of the paper is organized as follows: Section 2 highlights the traditional research based on DDoS attack detection and mitigation in SDN-IoT environment. Section 3 proposes the proposed model implementation in several phases. Section 4 outlines the experimental setup and discussions on results. Section 6 outlines conclusion and suggests areas for future investigation.

2. Overview of prior research

Prior research on DDoS attack detection in SDN-IoT networks included ML, DL and FL algorithms, each with advantages in accuracy and efficiency. However, limitations in flexibility, scalability, and dynamic deployment outline major gaps in research, which needs more robust solutions.

2.1. Machine learning and ensemble-based approaches

Belachew et al. [14] addressed IoT-SDN DDoS attacks using edge-deployed eXtreme Gradient Boosting (XGBoost), achieving high accuracy and minimal latency. However, employing benchmark datasets, restrict real world testing, and focusing on certain attacks shows gaps in research. Hirsi et al. [15] developed a supervised ML-based SDN traffic classifier utilizing a custom DDoS dataset, achieving impressive accuracy. However inadequate dataset diversity the possibility of overfitting and scalability issues highlight research shortcomings and restrictions on the robustness and generalizability of the method. Sinha et al. [16] discussed about DDoS-Blocker a lightweight SDN defense that prevents attacks using special features and random forest classification. However, relying on pre-configured features a lack of testing in various attack scenarios and potential scaling problems show limitations and draw attention to research gaps regarding adaptability and applicability. Khedr et al. [17] developed the Four-Module DDoS Attack Detection and Mitigation (FMDADM) an SDN-based four-module system for IoT DDoS detection that attains effective mitigation and high accuracy. However, the reliance on simulations, dataset limits, and restricted scalability evaluation indicated gaps in research adaptability, robustness, and generalizability. Alashhab et al. [18] developed an ensemble online ML model for SDN DDoS detection, which achieved high accuracy across various attack types. The system shows multiple essential shortcomings, which require further investigation because it depends on simulations and needs continuous retraining while it fails to effectively manage low-rate and zero-day attacks. Chaudhary et al. [19] developed a software-defined fog computing in IoT system, which consists of multiple stages that use entropy-based anomaly detection with dynamic thresholding and attribute selection ML.

2.2. Deep learning and hybrid architectures

Ahuja et al. [20] developed a hybrid DL model with a new SDN-DDoS dataset, which achieved precise results for detecting cyber-attacks. In contrast, limited diversity of data, dependence in experimental data and slow detection time highlight constraints and research requires model generalization and scalability. Zaidoun and Lachiri [21] proposed transformer and attention mechanism based a hybrid Deep Convolutional Neural Network – Long Short-Term Memory (DCNN-LSTM) to identify SDN DDoS. The research found that the system performance limits because it depended on the CIC-DDoS2019 dataset and used restricted datasets and had dynamic deployment capabilities. Wang and Wang [22] developed an SDN-based Low-rate DDoS (LDDoS) detection system which used Graph Convolutional Network with Gate Recurrent Unit (GCN-GRU) and OpenFlow to achieve accurate results while enabling immediate attack response. The need for defense systems which operate at full capacity and use minimal resources becomes clear through the risks which stem from using centralized controllers and the limited assessment of different IoT environments. Abid et al. [23] designed a CNN-LSTM multilevel deep neural network for DDoS detection in SDN-IoT, which captured spatial-temporal traffic aspects. However, the lack of large-scale validation high computational complexity and dependence on small real-world datasets exposed research gaps in deployment viability scalability and adaptability. Khan et al. [24] built weighted loop digraphs from nucleotide pairs to analyze DNA sequences using a graph theory-based alignment-free method. K-means clustering was used to assess sequence associations

and Cosine Correlation and Jaccard distance metrics are used to measure sequence similarity. In order to address the challenge of unbalanced datasets and effective feature selection, Yang et al. [25] suggested a reinforcement learning-based DDoS detection system using off-policy Proximal Policy Optimization (PPO). Despite these advances, existing solutions still face difficulties with scalability, real-time adaptability, dealing with imbalanced data and attaining high generalization performance in different and distributed IoT network settings.

2.3. Federated learning

Ge et al. [26] implemented a FL framework to extract local and global features from multimodal data, that combines RegNetZ and Swin-Transformer was developed. To improve convergence and scalability in decentralized environments a Hybrid Aquila–Grey Wolf Optimizer (HA-GWO) was utilized for effective hyperparameter tuning. Anjum et al. [27] developed a Graph-based Federated Artificial Intelligence (GraphFedAI) framework that combines Graph Neural Networks (GNNs) federated learning graph modelling based on adaptive session and attribute selection based on Pearson correlation to facilitate scalable and privacy aware analysis in IoT networks. The framework enhances feature extraction scalability and privacy in distributed environments by modelling IoT communication as dynamic graphs and training models locally. Aldrees et al. [28] address resource scarcity and communication backlogs in IoT networks a mutable resource allocation framework with distributed federated learning was proposed. The method makes it possible to use preference-based learning and adaptive edge-based resource management to boost network efficiency decrease allocation delays and improve connectivity. Wu [29] proposed Federated Averaging (FedAvg) and XGBoost based intrusion detection method with fog computing was proposed for wireless sensor networks, incorporating imbalanced data optimization and Top-K gradient selection to enhance detection accuracy, efficiency, and communication overhead. Wang et al. [30] implemented a FL-ResNeXt model was proposed for encrypted traffic classification, converting packet data into grayscale images and using an improved ResNeXt architecture to enhance feature extraction while preserving data privacy and improving classification accuracy. Izadi and Ahmadi [31] presented a cluster-based federated intrusion detection framework using a lightweight MobileNet model was proposed for heterogeneous IoT environments, where devices are grouped based on hardware similarity to improve scalability, reduce latency, and enhance privacy-preserving distributed learning. Ahanger et al. [32] proposed a federated learning-inspired framework for multi-class attack classification in IoT networks to improve detection while preserving data privacy. Talpini et al. [33] introduced a client clustering strategy in FL-based Intrusion Detection System(IDS) to enhance convergence and intrusion detection performance under heterogeneous IoT data distributions. Olanrewaju-George and Pranggono [34] developed an FL-based IDS combining unsupervised and supervised deep learning models to detect IoT intrusions effectively in distributed environments. Anastasakis et al. [35] presented a privacy-preserving FL-based IDS by integrating differential privacy to secure IoT systems against cyberattacks while mitigating data leakage risks. However, existing FL-based solutions face challenges in managing resources effectively, making flexible real-time decisions, and detecting attacks reliably in highly dynamic and diverse IoT settings. The current SDN-IoT DDoS detection and mitigation techniques are highlighted in this section along with their methodology and their significant advancements and constraints. Certain aspects of network security like accuracy, adaptability and distributed processing are addressed by a variety of approaches including traditional ML, DL, graph-based models, ensemble methods and FL frameworks. Despite these advancements, the majority of methods still have drawbacks including scalability issues, high computational and communication costs dependence on labeled or signature-based data model drift and challenges managing diverse and dynamic IoT environments.

The comparative analysis in Table 1 clearly demonstrates that existing SDN-IoT DDoS detection approaches mainly suffer from centralized dependency, computational overhead, scalability limitations, communication inefficiency, and poor adaptability to dynamic attack environments. Unlike this, the proposed FedMP-LDL framework adopts a lightweight multi-phase architecture, which combines Hamacher FERB based anomaly filtering, progressively distilled Ghost-SE model with FL, and reinforcement learning based dynamic SDN flow adaptation. The proposed framework provides efficient distributed learning, low computational complexity, high scalability, privacy protection, and adaptive real-time DDoS mitigation in heterogeneous SDN-fog IoT environments, which are not provided by existing approaches.

2.4. Problem statement

The current SDN-fog-IoT DDoS detection and mitigation solutions have many unaddressed limitations, such as high computation overhead, very complex models, growing communication costs in FL, and sub-optimal adaptation to changing and zero-day DDoS attacks. Most previous solutions rely on centralized training or computationally intensive DL architectures, which raises inference latency and renders them unsuitable for real-time fog-based deployment. Furthermore, current federated SDN-IoT security frameworks struggle with non-IID distributed traffic patterns, resulting in sluggish convergence and limited scalability. To address these issues, this research proposes a multi-phase lightweight framework that enables efficient detection and mitigation by combining fast rule-based filtering capable of rejecting benign traffic within 0.01 ms, dimensionality reduction achieving feature reduction and limiting the model to only 2.1M parameters, lightweight Ghost-SE attention providing faster inference with only low prediction time, adaptive optimization improving convergence by and completing training in only 100 communication rounds, and reinforcement learning-based SDN flow adaptation to handle dynamic and zero-day DDoS attacks effectively.

Table 1
Comparative analysis of existing SDN-IoT DDoS detection and FL-based methods with their specific drawbacks and key improvements of the proposed framework.

Existing method	Technique	Drawbacks	Key improvements in proposed FedMP-LDL framework
[14]	Edge-deployed XGBoost for IoT-SDN DDoS detection	Uses benchmark datasets only, lacks real-world deployment validation, and not effectively detect unknown attack patterns.	FedMP-LDL enables adaptive FL with real-time distributed detection for improved generalization under varying attack patterns.
[15]	Supervised ML-based SDN traffic classifier	Suffers from overfitting, poor scalability with large traffic volumes, and limited dataset diversity.	Hierarchical FL improves scalability and reduces overfitting through distributed learning across fog nodes.
[16]	DDoSBlocker using RF	Depends on manually engineered features and performs poorly under dynamic attack variations.	KPCA-TCN Transformer automatically extracts temporal traffic features and adapts to dynamic attack behavior.
[17]	FMDADM four-module SDN framework	Simulation-only evaluation and limited scalability testing reduce practical applicability.	FedMP-LDL supports scalable deployment in distributed SDN-fog IoT environments with lightweight inference capability.
[18]	Ensemble online ML model	Requires continuous retraining and has weak detection capability for zero-day and low-rate attacks.	RL-driven adaptive flow rule optimization improves zero-day and varying attack mitigation without frequent retraining.
[19]	Entropy-based anomaly detection with ML	Dynamic threshold selection is unstable under fluctuating IoT traffic conditions.	Hamacher FERB provides uncertainty-aware anomaly filtering with stable adaptive decision-making.
[20]	Hybrid DL model	Slow detection speed and poor generalization due to limited training diversity.	Lightweight Ghost-SE architecture reduces inference time while improving detection accuracy and generalization.
[21]	Transformer-based DCNN-LSTM	Strong dependence on CIC-DDoS2019 dataset and high computational complexity.	Progressive Ghost-SE distillation significantly reduces model parameters and computational overhead.
[22]	GCN-GRU with OpenFlow	Centralized SDN controller creates a single point of failure and increases resource overhead.	Distributed federated architecture minimizes centralized dependency and enhances scalability.
[23]	CNN-LSTM multilevel network	High computational cost and lack of validation in large-scale real-time IoT environments.	Lightweight multi-phase architecture supports efficient real-time deployment on resource-limited fog nodes.
[24]	Graph theory-based sequence analysis	Not suitable for real-time SDN-IoT traffic analysis and attack mitigation.	FedMP-LDL is specifically designed for real-time SDN-fog IoT DDoS detection and adaptive mitigation.
[25]	PPO-based reinforcement learning	Requires extensive training time and struggles with highly imbalanced datasets.	Multi-objective Q-learning improves adaptive mitigation with lower training complexity and faster decision-making.
[26]	FL with RegNetZ and Swin-transformer	High communication overhead and increased training complexity in decentralized environments.	H-FedAvg reduces communication overhead and supports efficient federated aggregation.
[27]	GraphFedAI using GNN and FL	Graph construction complexity and scalability issues in heterogeneous IoT networks.	FedMP-LDL avoids excessive graph construction complexity while maintaining scalability across heterogeneous IoT devices.
[28]	Distributed FL resource allocation framework	Resource allocation delay and communication bottlenecks under dense IoT traffic.	RL-DFRA dynamically optimizes SDN flow adaptation and reduces network congestion during attacks.
[29]	FedAvg and XGBoost-based IDS	Gradient exchange increases communication cost and weakens adaptability to evolving attacks.	Lightweight hierarchical FL minimizes communication overhead and improves adaptive learning.
[30]	FL-ResNeXt encrypted traffic classification	Image conversion process increases computational burden and memory usage.	FedMP-LDL directly processes traffic features without costly image transformation operations.
[31]	Cluster-based FL IDS using MobileNet	Hardware-based clustering reduces flexibility in highly dynamic IoT systems.	Adaptive multi-phase FL improves flexibility under dynamic SDN-IoT environments.
[32]	FL-inspired multi-class attack classification	Limited adaptability to rapidly changing attack behaviors.	RL-driven adaptive mitigation continuously updates flow rules according to attack intensity variations.
[33]	Client clustering in FL-based IDS	Client heterogeneity negatively affects convergence stability and detection consistency.	H-FedAvg enhances convergence stability across heterogeneous fog nodes and IoT devices.
[34]	FL-based IDS with supervised and unsupervised DL	Hybrid model increases computational complexity and communication overhead.	Ghost-SE lightweight architecture significantly reduces complexity while maintaining high detection accuracy.
[35]	Differential privacy-based FL IDS	Privacy mechanisms reduce detection accuracy and increase model training complexity.	FedMP-LDL preserves privacy through FL while maintaining high detection accuracy and low latency.

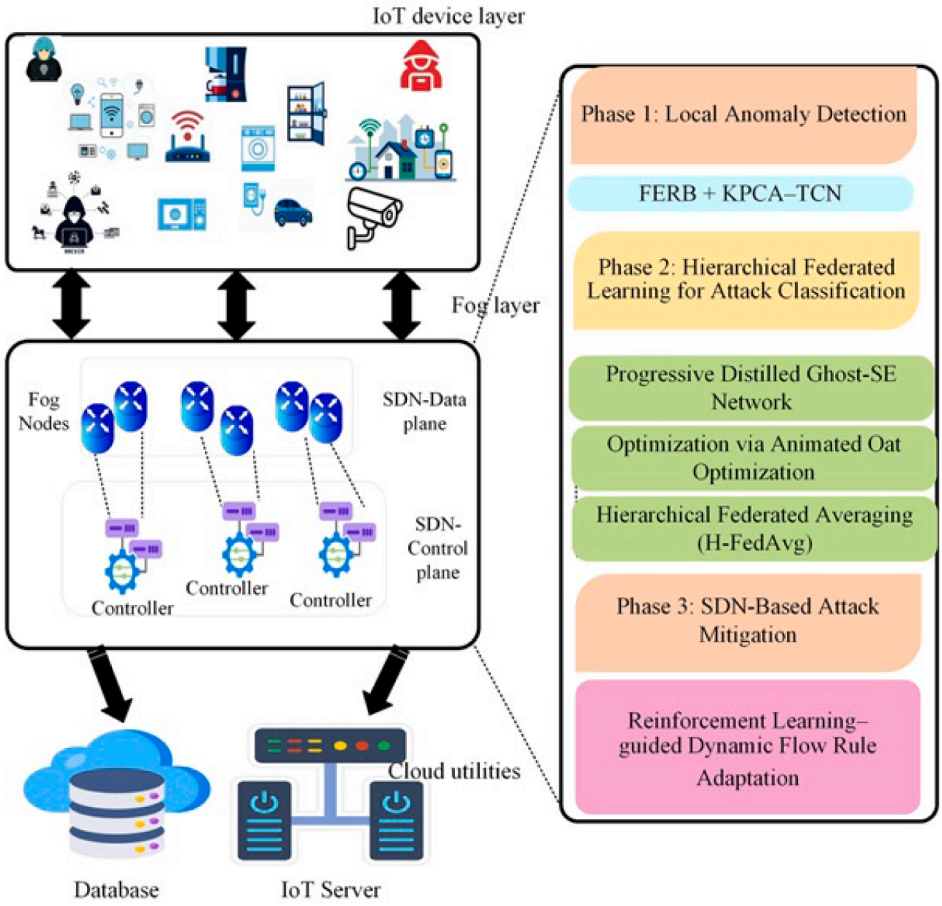


Fig. 1. Proposed federated multi-phase lightweight deep learning architecture.

3. Proposed methodology of attack detection and mitigation in SDN-fog-enabled IoT environments

The proposed FedMP-LDL architecture is implemented to facilitate scalable, privacy preserving and efficient DDoS detection and mitigation in SDN-fog-enabled IoT environments, depicted in Fig. 1. The proposed architecture processing pipeline includes traffic monitoring, local anomaly detection, federated classification, and SDN-based mitigation, followed by performance assessment.

3.1. Traffic monitoring and preprocessing

Data discretization and normalization techniques are used to preprocess the raw IoT traffic data at each fog node, to improve detection performance. The discretization preprocessing technique is used to transform continuous in to limited set of representative values, which improve interpretability and eliminate noise. Normalization is used to scale these attributes into the specific range value [0,1] using Eq. (1).

$$N(z) = \frac{z - z_{\min}}{z_{\max} - z_{\min}} \tag{1}$$

where, z denoted as feature value, and z_{min}, z_{max} defines minimum and maximum values of features. This preprocessing pipeline facilitate well scaled and compact feature sets suitable for lightweight anomaly detection and federated classification.

3.2. Phase 1 – local anomaly detection

In the proposed FedMP-LDL architecture, FERB based preliminary anomaly detection is processed at the fog layer, which is followed by a sequential detector KPCA–TCN for suspicious flows detection. This phase reduces computational overhead by filtering 92% benign traffic before deep learning classification. The FERB model replaces strict threshold rules with Hamacher fuzzy equivalence relations, allowing gradual transitions near decision boundaries to be represented [36]. Instead of crisp cutoffs, packet rate and flow duration are mapped to fuzzy membership values that quantify their similarity to reference thresholds.

These similarity measures are then aggregated into a single suspicion score, which determines whether traffic is anomalous and forwarded for deeper analysis. For example, the strict rule

If (PacketRate $> \phi_r) AND$ (FlowDuration $< \phi_d$) then Anomaly = TRUEis replaced with the fuzzy rule

If (PacketRate $\approx \phi_r) AND$ (FlowDuration $\approx \phi_d$) then

SuspicionScore = High

where, $\approx$ indicates partial similarity rather than an exact threshold match, ϕ_r is the reference threshold for packet rate represents the boundary above which traffic is considered suspicious in terms of number of packets per second are being sent. ϕ_d the reference threshold for flow duration represents the boundary below which traffic is considered suspicious in terms of short-lived a TCP/flow connection performance. The similarity of observed values to thresholds is quantified through fuzzy equivalence relations, defined as in Eq. (2).

$$E(y, z) : Y \times Y \rightarrow [0, 1] \tag{2}$$

where $E(y, z)$ denotes the degree of similarity between two valuesyand z, and the range [0,1] indicates complete dissimilarity to full similarity. Using Hamacher's equivalence, the similarity is expressed as in Eq. (3).

$$E(y, z) = \frac{\eta}{e^{d(y,z)} - 1 + \eta} \tag{3}$$

where $dev(y, z)$ represents the absolute deviation of the observed feature value from the reference threshold, and $\eta \neq 0$ is a tunable parameter optimized empirically. Equivalence scores are separately computed for packet rate and flow duration, denoted as E_r, E_d. These scores are aggregated using a parameterized operator defined in Eq. (4).

$$E_{\text{agg}} = \frac{\eta}{e^{(p_e d_r + p_d d_d)} - 1 + \eta} \tag{4}$$

where p_r and p_d represent weights associated with the relative importance of packet rate and flow duration, d_r and d_d is deviations. Finally, the suspicion score is $S_{anom} = 1 - E_{agg}$, which maps the similarity measure into an anomaly likelihood within range $[0, 1]$. A higher value of S_{anom} indicates stronger deviation from normal behaviour. Only traffic with high suspicion scores is forwarded to the KPCA-TCN model for deeper analysis, while benign traffic is filtered at the fog level. Algorithm 1 demonstrates the processing steps for FERB based local anomaly detection.

Algorithm 1 FedMP-LDL_local_anomaly_detection.

Input:
Traffic Flows = $\{f_1, f_2, \ldots, f_n\}$
Thresholds: ϕ_r (packet rate), ϕ_d (flow duration)
Parameters: η (Hamacher parameter), p_r, p_d (feature weights)
Output:
Suspicious Flows (subset of Traffic Flows)
Begin
Initialize Suspicious Flows
For each flow f in Traffic Flows do
Step 1: $d_r = |\text{packet rate}(f) - \phi_r|$
$d_d = |\text{flow duration}(f) - \phi_d|$
Step 2: Compute similarity using Hamacher equivalence
$E(y, z) = \frac{\eta}{e^{d(y,z)} - 1 + \eta}$
Step 3: Aggregate similarities
$E_{agg} = \frac{\eta}{e^{(p_r d_r + p_d d_d)} - 1 + \eta}$
Step 4: Compute suspicion score
$S_{anom} = 1 - E_{agg}, \quad S_{anom} \in [0, 1]$
Step 5: Decision rule
If S_{anom} is high
add f to Suspicious Flows
End If
End For
Step 6: Forward suspicious flows for deeper analysis
For each flow fin Suspcious Flows do
Forward to KPCA-TCN_Sequential Detector
End For
End

The novelty of the FERB local anomaly detection lies in its pioneering use of Hamacher equivalence relations applied for the first time to DDoS pre-filtering to replace rigid threshold-based rules with smooth partial similarity scoring of packet rate and flow

duration against reference attack boundaries. Unlike conventional fuzzy systems with fixed membership functions, FERB incorporates learnable weighted aggregation and an intuitive suspicion inversion mechanism, enabling more effective benign traffic filtering while maintaining high anomaly detection accuracy, making it well-suited for efficient deployment at fog nodes.

3.2.1. KPCA–TCN sequential detection

FERB detected suspicious flows are forwarded to a KPCA -TCA module in order to eliminate redundant features and to capture temporal dependencies in network traffic. This two-step detection ensures that only truly anomalous flows proceed to the federated classification phase, optimizing computational efficiency at the fog node.

KPCA is applied to the normalized traffic feature matrix $Y \in \Gamma^{m \times n}$, where mis number of samples and n is number of network features [37]. The main idea is to project non-linear data into high-dimensional feature space via a mapping $\phi : \Gamma^n \rightarrow \Gamma^h \quad (h \gg n)$, where linear PCA is applied by Eq. (5).

$$\phi(y_i)^T \phi(y_j) = k(y_i, y_j) = K_{ij} \tag{5}$$

where, $k(y_i, y_j)$ is a kernel function. The polynomial kernel is used within the KPCA module to capture non-linear relationships in network traffic data. It is defined in Eq. (6).

$$k(y_i, y_j) = (y_i \cdot y_j + 1)^{\deg} \tag{6}$$

where y_i and y_j are traffic feature vectors extracted at fog nodes. $y_i \cdot y_j$ represents the similarity between two traffic samples in the original feature space. deg is the degree of the polynomial, controlling the flexibility of the projection: higher degcaptures more complex non-linear dependencies in traffic behaviour. The centred kernel matrix $tilde\Psi$ in KPCA is used to ensure that the data in the high-dimensional feature space has zero mean, which is a key requirement for PCA, computed by using Eq. (7)

$$\tilde{\Psi} = \Psi - \mathbf{1}_m \Psi - \Psi \mathbf{1}_m + \mathbf{1}_m \Psi \mathbf{1}_m \tag{7}$$

where, $\mathbf{1}_m = \mathbf{1}\mathbf{1}^T$ is an $m \cdot m$ matrix of ones, scaled by $\frac{1}{n}$. After centering, KPCA solves the eigenvalue problem by using Eq. (8).

$$\tilde{\Psi}\, \beta = \eta \beta \tag{8}$$

where, β is the eigenvector in the kernel space. η is the corresponding eigenvalue. Each eigenvector β_j corresponds to a principal component direction in the high-dimensional space. Once the eigenvectors and eigenvalues η are obtained, the principal components u_j in the feature space are computed as in Eq. (9).

$$u_j = \sum_{i=1}^{m} \beta_i^{(j)} \phi(y_i), \quad j = 1, 2, 3, \ldots, l \tag{9}$$

where, l is the number of components retained typically much smaller than m to reduce dimensionality. Each u_j represents a direction in feature space capturing maximal variance in the traffic data. The low-dimensional representation Tof the original data is then defined by Eq. (10).

$$T = [t_1, t_2, t_3, \ldots, t_l]^T, \quad t_j = \phi(Y)^T u_j \tag{10}$$

where, t_j are the principal component scores for each sample. These scores form a compact, informative feature set suitable for temporal anomaly detection. KPCA allows up to m principal components, matching the number of traffic samples. For each new traffic observation, its projected representation in the feature space is computed, which serves as input for the anomaly detection unit. TCN captures temporal dependencies in KPCA-transformed sequences $T = \{t_1, t_2, \ldots, t_l\}$ using causal and dilated convolutions [38]. Causal convolution ensures that output at time t depends only on previous inputs, preventing information leakage from future traffic. Dilated convolution expands the receptive field without increasing parameters, enabling long-range dependency modelling in IoT traffic sequences using Eq. (11).

$$z[t] = \sum_{i=0}^{k-1} f[i] \cdot y[t - d \cdot i] \tag{11}$$

where, $f[i]$ signifies convolution kernel, k denotes kernel size, and d referred to dilation factor. To improve performance as network depth increases, residual connections are integrated to the model, expressed in Eq. (12).

$$r(Y) = \text{Activation}\big(\Xi(Y) + Y\big) \tag{12}$$

where, $\Xi(Y)$ denotes residual function and $Activation()$ represents a non-linear function, which prevents vanishing gradients, maintains learning efficiency and facilitates stable gradient flow in deep TCN layers. By leveraging KPCA model for dimensionality reduction and TCN model for capturing temporal dependencies, the architecture effectively maintains critical traffic patterns and facilitating accurate, lightweight anomaly detection at fog nodes. This sequential detector module assures that both spatial and temporal features are efficiently leveraged for detecting DDoS and other network anomalies in resource-constrained IoT environments.

The novelty of the KPCA-TCN sequential detection resides in its innovative two-stage architecture, which integrates polynomial kernel-based KPCA for non-linear dimensionality reduction with causal-dilated Temporal Convolutional Network modeling for temporal dependency extraction, the first documented application of this combination for fog-based DDoS anomaly confirmation in SDN-IoT environments.

3.3. Phase 2- hierarchical federated learning for attack classification

Filtered traffic is processed using Distilled Ghost-SE Network which integrates a progressive distillation strategy with the GhostNet architecture and SE modules. This compact DL model facilitates high-accuracy in attack classification and minimize computational overhead. Algorithm 2 demonstrates the processing steps for hierarchical federated learning for attack classification. The novelty of the Ghost-SE Network with H-FedAvg training resides in its pioneering integration of GhostNet bottleneck efficiency, Squeeze-and-Excitation channel recalibration, progressive knowledge distillation, and Animated Oat Optimization (AOO) within a hierarchical federated averaging (H-FedAvg) framework, the approach enabling fog-based DDoS classification with 67% parameter reduction, 2.3 times faster convergence, and privacy-preserving scalability across heterogeneous IoT–Fog–SDN architectures.

3.3.1. Preserving feature representation using the GhostNet module

GhostNet model is implemented to generate rich feature maps by using a standard convolutional functions and linear operations, in which the feature representation quality is preserved and substantially computational overhead is reduced [39]. KPCA based reduced features and TCN based extracted temporal patterns The GhostNet module processes IoT traffic feature vectors, in terms of spatial features and temporal patterns, to detect discriminative traffic pattern representations suitable for local attack classification. Each bottleneck block of GhostNet includes two Ghost modules, which are arranged in a sequential manner. The first module uses standard convolution to create a base set of intrinsic feature maps, which generated by expanding the input feature space. The second module uses low-cost linear operations, which include depth-wise convolution and channel-wise transformations, to create additional ghost feature maps that effectively capture redundant patterns. The compressed feature vector for each traffic observation is then obtained by combining the outputs of these two modules defined in Eq. (13).

$$Fe_{\text{node}} = Fe_{\text{conv}} + Fe_{\text{linear}} \tag{13}$$

where, Fe_{conv} represents obtained features by using standard convolution, while Fe_{linear} denotes derived features by using linear transformations such as depth wise convolution, average pooling, or channel-wise operations. The number of output channels allotted to each node is used to calculate the final feature dimension. In terms of stride and residual connections the Ghost bottleneck maintains feature consistency by aligning output channels using shortcuts in a residual design for stride $= 1$. Inspired by the linear bottleneck module of MobileNetV2, a down sampling operation is used for stride $= 2$ in order to modify spatial dimensions. After each convolution or linear operation Batch Normalization (BN) and ReLU activation are helpful to guarantee model stability and non-linearity with the exception of the final block where ReLU is skipped to maintain feature fidelity for further processing.

3.3.2. Squeeze-and-excitation blocks for attack feature emphasis

The SE block enhances the sensitivity of convolutional features to informative signals by explicitly modelling channel interdependencies [39]. This mechanism enables the model to focus on relevant attack characteristics rather than irrelevant signals. For instance, in the case of a scale attack, which produces a sudden amplitude jump compared to normal traffic, the SE block assigns a higher weight to this jump feature, allowing the model to effectively distinguish it. The SE block operates in two steps: squeezing and excitation. In squeezing step, global spatial information from each feature channel is compressed into single channel descriptor using global average pooling. Mathematically, expressed as using Eq. (14).

$$s = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} v(i,j) \tag{14}$$

where H and W represent height and width of features from previous layer, and $v(i,j)$ denotes data element in feature map. For one-dimensional convolution, width is considered $W = 1$. If the SE block is applied after a concatenation operation, $v(i,j) \in P_i$. In the excitation step, the channel interdependencies are adaptively reweighted using a gating mechanism, defined as in Eq. (15).

$$e = v(i,j) \cdot \gamma\big(g(s,W)\big) = v(i,j) \cdot \gamma\big(W_1 \gamma(s, W_2)\big) \tag{15}$$

where, W_1 and W_2 represent the learnable weights, and γ denotes the ReLU activation function. This operation multiplies each input data element by its corresponding channel factor, effectively acting as a self-attention mechanism across all channel features. By emphasizing the most informative channels, the SE block improves model's capability to detect and classify DDoS attack patterns accurately.

In order to create a lightweight student network progressive distillation iteratively compresses knowledge from a larger teacher model [40]. The student is trained on noisy traffic representations at each iteration after being initialized with the teachers' parameters. In order to ensure sharper and more informative representations the student predicts intermediate targets that are derived from the teacher's deeper inference steps rather than directly learning from raw labels. The student model gradually learns to approximate the teacher's performance with fewer computations by halving the number of sampling steps at each iteration. Fog nodes carry out precise resource-efficient attack classification locally by combining Ghost-SE enhancements with progressive knowledge distillation. Algorithm 3 shows the processing steps for Ghost-SE network's progressive distillation.

3.3.3. Training optimization with animated oat optimization (AOO)

The AOO algorithm is a bio-inspired metaheuristic based on the dispersal behavior of animated oat seeds [41]. In this research, AOO is used to optimize the training of the Distilled Ghost-SE Network, accelerating convergence, maintaining model compactness,

Algorithm 2 Hierarchical federated learning for attack classification.

Input:

- IoT clients $\{C_1, C_2, \ldots, C_n\}$
- Fog nodes $\{F_1, F_2, \ldots, F_m\}$
- SDN controller (global aggregator)
- Filtered data D_i (from Phase 1 – FERB)
- Learning rate μ, communication rounds T
- AOO parameters $(L_b, U_b, Dim, T_{iter})$

Output:

- Final global model w^T

Initialization

Step 1: Initialize global model w^0 (Ghost-SE Network)
Step 2: Initialize AOO candidate solutions using Eq. (16):
$z_{(i,j)} = \text{rand} \times (Ub_j - Lb_j) + Lb_j$
for each communication round $t = 0$ to $T - 1$ **do**
Stage 1 Local Training at IoT Clients
Step 3: For each IoT client C_i (in parallel)
Step 4: Receive fog-level model w_f^t
Step 5: Apply Ghost feature extraction (Eq. 13):
$F_{node} = F_{conv} + F_{linear}$
Step 6: Apply SE squeeze (Eq. 14):
$s = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x(i,j)$
Step 7: Apply SE excitation (Eq. 15):
$e = x(i,j) \cdot \gamma(W_1\, r(s, W_2))$
Step 8: Perform progressive knowledge distillation
Initialize student from global model
Learn intermediate representations
Step 9: Update local model using gradient descent:
$w_i^{(t+1)} = w_f^t - \mu \cdot \nabla l(w_f^t; D_i)$
Step 10: Send $w_i^{(t+1)}$ to associated fog node
Stage 2: Fog-Level Aggregation + AOO Optimization
Step 11: For each fog node F_f
Collect updates $\{w_i^{(t+1)}\}$
Step 12: Compute weighted aggregation (H-FedAvg):
$w_f^{(t+1)} = \frac{\sum |D_i| \cdot w_i^{(t+1)}}{\sum |D_i|}$
Step 13: Apply AOO parameter dynamics (Eq. 17):
$s_m, a_{len}, e_r, d_{y_{ad}} = f(\text{rand}_{Dim}, t, T_{iter})$
Step 14: Update candidate solutions using Eq. (18):
$J = \frac{2kz^2 \sin(2\phi)}{s_m \cdot \text{rand}_{Dim}(B, B_s) \cdot (1-\beta)}$
Step 15: Perform AOO-based outlier filtering
Remove anomalous or poisoned updates
Step 16: Send optimized fog model $w_f^{(t+1)}$ to SDN controller
Stage 3: Global Aggregation at SDN Controller
Step 17: Collect $\{w_f^{(t+1)}\}$ from all fog nodes
Step 18: Compute global aggregation:
$w^{(t+1)} = \frac{\sum |D_f| \cdot w_f^{(t+1)}}{\sum |D_f|}$
Step 19: Broadcast updated global model $w^{(t+1)}$
end for
if convergence criteria satisfied **then**
Step 20: terminate
end if
Step 21: Output final model w^T

Algorithm 3 Progressive distillation.

Require:

- Trained teacher model f_η (e.g., Ghost-SE teacher)
- Filtered traffic data
- Weight loss function $\omega(\cdot)$
- Student sampling steps S

for J iteration **do**
  Initialize student from teacher: $\phi \leftarrow \eta$
  while not converged **do**
    Sample traffic flow $y \sim$ Data
    Add perturbation:
    $x_t = \beta_t x + \sigma_t \epsilon,\ \epsilon \sim \mathcal{N}(0, I)$
    Teacher performs 2-step inference:
    $x'_t = f_\eta(x_t)$
    $x''_t = f_\eta(x'_t)$
    Define distilled target:
    $\tilde{z} = \text{Distil}(x_t, x''_t)$
    Student prediction (Ghost-SE backbone):
    $\hat{z}_\phi = f_\phi^{(\text{Ghost-SE})}(x_t)$
    Compute loss:
    $\mathcal{L}_\phi = \omega\left(\lambda_t \left\| \tilde{z} - \hat{z}_\phi \right\|^2\right)$
    Update student:
    $\phi \leftarrow \phi - \gamma \nabla_\phi \mathcal{L}_\phi$
  end while
  Update teacher:
  $\eta \leftarrow \phi$
  Reduce sampling steps:
  $S \leftarrow S/2$
end for

and improving resource-efficient classification at fog nodes. The behavior of AOO is inspired by the natural dispersal strategies of animated oat seeds, with two key survival mechanisms shown in Fig. 2

AOO begins by generating a set of random candidate solutions for the model parameters, z_i represents the ith candidate solution (individual) with Dim dimensions, and M denotes the population size. Each element $z_{(i,j)}$ is initialized as defined in Eq. (16).

$$z_{(i,j)} = \text{rand} \times (Ub_j - Lb_j) + Lb_j, \quad i = 1, \ldots, \text{Dim} \tag{16}$$

where $\text{rand} \in [0, 1]$ is random number, and Ub_j, Lb_j are upper and lower bounds of the jth parameter.

Key parameters guiding the animated oat seeds' movement are derived from seed morphology and dynamics defined in Eq. (17).

$$s_m = 0.5 \times \text{rand}_{\text{Dim}}, \quad A_{\text{len}} = M \times \text{rand}_{\text{Dim}}, \quad e_r = 0.5 \times \text{rand}_{\text{Dim}}, \quad d_{y_{ad}} = 1 - \left(\frac{t}{T}\right)^3 \tag{17}$$

where, s_{mis} seed mass, A_{len} denotes main awn length, e_{ris} the eccentric rotation coefficient, dy_{ad} denotes dynamic adjustment factor, t current iteration, Tmaximum iterations, and $rand_{Dim}$ random number per dimension. dy_ad controls the transition from global exploration (random dispersal with diverse updates) to local exploitation (refinement around promising regions). During exploration, candidate solutions are perturbed using stochastic displacement and multi-strategy updates (population mean, best solution guidance, and random motion), ensuring diversity and global search capability. In the exploitation phase, solutions are refined using local adaptive dynamics combined with Lévy flight-based perturbations, enabling escape from local optima and improving convergence stability. To enhance convergence, a dynamic boundary contraction and stochastic motion mechanism is applied. The final position update is expressed as, in Eq. (18).

$$J = \frac{2kz^2 \sin(2\phi)}{s_m \times \text{rand}_{\text{Dim}}(-B, B)(1 - \beta)} \tag{18}$$

where, B represents the adaptive boundary, B_s is the lower bound, β controls exploration–exploitation balance, and k is a scaling factor.

The proposed AOO-based optimization effectively enhances convergence speed, model compactness (67% parameter reduction), and resource efficiency for fog-based Ghost-SE classification. Its adaptive exploration-exploitation mechanism, mimicking oat seed

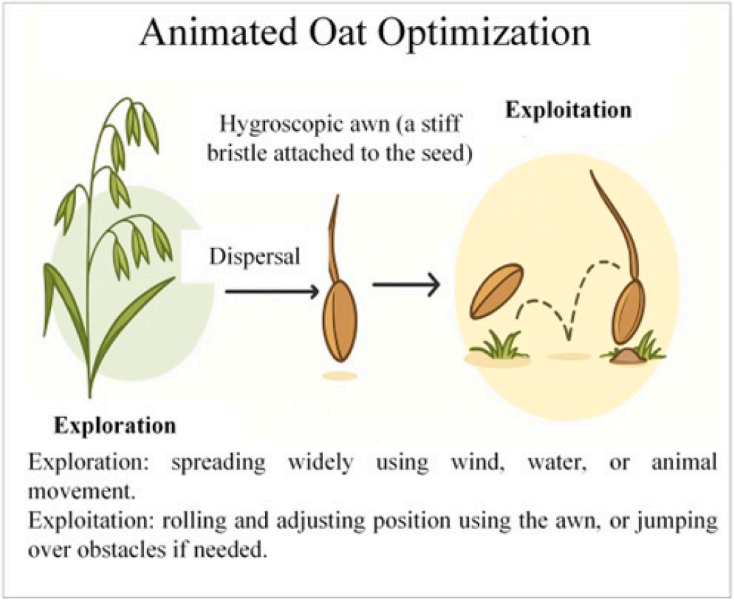


Fig. 2. Behavior of AOO.

dispersal physics ensures robust performance across heterogeneous IoT clients while maintaining low computational overhead suitable for edge deployment.

3.3.4. *Hierarchical federated averaging for global training*

The proposed FedMP-LDL framework needs to fit the three-tiered IoT–Fog–SDN architecture. Standard FedAvg is not suitable since it causes excessive communication overhead and latency if there are many IoT devices communicating directly with the SDN controller (global aggregator) [42]. In order to overcome this drawback, the framework uses H-FedAvg, which makes fog nodes as intermediate aggregators between the IoT clients and the SDN controller. The learning process takes place in three phases:

Local Training at IoT Clients: Each IoT device trains its lightweight Distilled Ghost-SE model on locally observed traffic, optimized using the AOO algorithm. For a given IoT client i with dataset Dat_i, the model update at round t is expressed as in Eq. (19).

$$w_i^{(t+1)} = w^t - \mu \nabla l\left(w^t; \mathrm{Dat}_i\right) \tag{19}$$

where μ is the learning rate and $l(\cdot)$ represents the local loss function.

Fog-Level Aggregation: Fog nodes act as intermediate aggregators, collecting model updates from their associated IoT devices and producing refined sub-global models. For a fog node fwith client set $Client_f$,the aggregation is defined as in Eq. (20).

$$w_f^{(t+1)} = \frac{\sum_{i \in \mathrm{Client}_f} \left|\mathrm{Dat}_i\right| w_i^{(t+1)}}{\sum_{i \in \mathrm{Client}_f} \left|\mathrm{Dat}_i\right|} \tag{20}$$

This weighted averaging smooths noisy updates and reduces communication overhead before sending results to the SDN controller.

Global Aggregation at SDN Controller: The refined models from fog nodes are sent to the SDN controller, where H-FedAvg combines them into a global model that captures diverse attack patterns across the entire network. The controller then redistributes this global model back to fog nodes and IoT devices. The SDN controller collects sub-global models from all fog nodes F_{node} and performs global aggregation defined in Eq. (21).

$$w_g^{(t+1)} = \frac{\sum_{f \in F_{\mathrm{Node}}} \left|\mathrm{Dat}_f\right| w_f^{(t+1)}}{\sum_{f \in F_{\mathrm{Node}}} \left|\mathrm{Dat}_f\right|} \tag{21}$$

where $\left|\mathrm{Dat}_f\right|$ is the total data size managed by the fog node f. The updated global model $w_g^{(t+1)}$ is redistributed back to fog nodes and IoT clients. Through this hierarchical design, the framework achieves efficient, privacy-preserving, and scalable attack classification. By collaboratively training across IoT clients, fog nodes, and the SDN controller, the global model effectively captures heterogeneous attack behaviors while minimizing latency and communication costs.

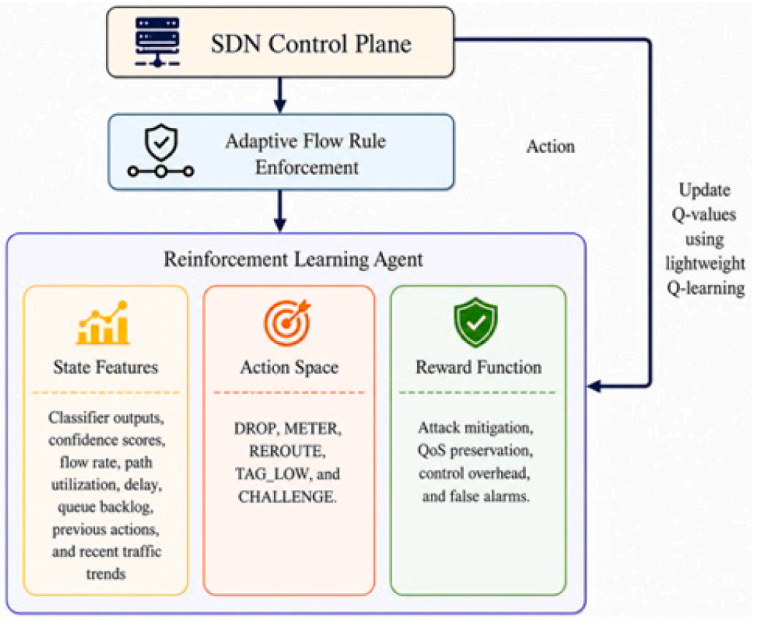


Fig. 3. Reinforcement learning–guided dynamic flow rule adaptation for mitigation.

3.4. Phase 3- SDN-based attack mitigation

In the third phase, the global classifier model produced by hierarchical federated learning is deployed across fog nodes and integrated with the SDN control plane. The SDN controller employs an RL-DFRA framework to mitigate detected attacks, illustrated in Fig. 3.

The primary objective of RL-DFRA is to enforce adaptive and context-aware flow control decisions that suppress malicious activity while preserving benign traffic performance and minimizing flow-table overhead [43]. Consider the SDN–IoT network as a graph $Graph = (N, E)$ with switches $N = \{N_i\}$ and links $E = \{l_{(i,j)}\}$. Each switch N_i with a flow-table occupancy F_i and maximum capacity $F_i^{\max}$, while each link l_{ij} with an utilization C_{ij} and capacity $C_{ij}^{\max}$. End-to-end delay for path pis $l_p = \sum_{l \in p} D_l$, constrained by $l_p \le l_{\max}$. Path cost for a candidate forwarding path P_n is expressed as in Eq. (22).

$$\Xi_n = \sum_{N_i \in \mathbb{W}_n} \left(\frac{F_i}{F_i^{\max}} + \frac{1}{\phi_i} \right) + \sum_{l_i \in L_n} \left(\frac{C_{ij}}{C_{ij}^{\max}} + D_{ij} \right) \tag{22}$$

where W_n and L_n denote the switch and link sets along P_n. Within this environment, the RL agent operates at discrete control epochs t. The state space is defined for each candidate flow f_c as defined in Eq. (23).

$$s_t = \left[\hat{y}_{f_c},\ p_{f_c},\ r_{f_c},\ u_{\text{path}(f_c)},\ l_{f_c},\ \frac{F_i}{\left(F_i^{\max}\right)_{\text{egres},\,t-1,\,f_c}} \right] \tag{23}$$

where $\hat{y}_{f_c}$ and p_{f_c} represent classifier output and confidence, r_{f_c} the flow rate, $u_{\text{path}(f_c)}$ path utilization, l_{f_c} delay, q_{egress} queue backlog, $a_{(t-1)}$ the previous action, and Δr_{f_c} the recent traffic trend. The action space is a discrete set with optional parameters defined in Eq. (24).

$$\mathcal{A} = \{\text{ALLOW}, \text{TAG_FLOW}, \text{METER}(\rho, \beta), \text{DROP}, \text{REROUTE}(k), \text{CHALLENGE}\} \tag{24}$$

Actions map directly to SDN flow operations: e.g., DROP inserts a high-priority drop rule, METER applies rate limiting with parameters (ρ, β), REROUTE selects an alternate path minimizing Ξ_n, TAG_LOW reduces scheduling priority, and CHALLENGE redirects flows to verification services. The reward function balances attack mitigation, QoS, and control cost defined in Eq. (25).

$$Rew_t = \alpha_1 \Delta U_{\text{benign}} - \alpha_2 L_{\text{benign}} + \alpha_3 B_{\text{blocked}} - \alpha_4 F A_t - \alpha_5 C_{\text{ctrl}} - \alpha_6 \Delta l_{\text{prio}} \tag{25}$$

where ΔU_{benign} denotes change in benign throughput, L_{benign} loss, $B_{blocked}$ malicious bytes blocked, $F_{A,i}$ false alarms, $Cctrl$ control overhead (rule churn, flow updates), and Δl_{prio} additional delay for priority flows. Weighting factors α_i allow policy tuning, typically with $\alpha_2 > \alpha_3$ to emphasize QoS preservation. Learning is achieved through a lightweight Q-learning update using Eq. (26).

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \mu \left[\text{Rew}_t + \eta \max Q(s_{t+1}, a') - Q(s_t, a_t) \right] \tag{26}$$

where, μ denotes learning rate and η represents discount factor. Exploration is governed by an ε -greedy policy wherein an action maximizing $Q(s_t, a)$ is chosen if there is a possibility of ε random action. The state features are discretized and normalized and

Table 2
Hardware and software configuration for experimental setup.

Component	Specifications
Fog Nodes	CPU: Intel® Core™i5-6600k, 3.9 GHz RAM: 16 GB Storage: 1 TB HDD OS: Ubuntu
Cloud Server	RAM: 32 GB Storage: 2 TB HDD
Victim Server	CPU: Intel® Core™i3-6700k, 3.2 GHz RAM: 4 GB Storage: 500 GB HDD

Table 3
Network topology configuration.

Components	Configuration Details
SDN Protocol	OpenFlow v1.5.1
Controller	SDN Controller: ONOS 2.7.0 (Ryu v4.34 fallback), hosted on Fog Node (PC2)
Network Scale	4x Open vSwitch 2.17.0 (S1–S4), 20 IoT Devices: Raspberry Pi 4 (2GB RAM) (D1–D20)
Connections	100 Mbps with realistic delays (5–25 ms RTT)
Malicious Actors	4 Malicious Devices (D5, D6, D10, D11)
Attack Tool	hping3 (TCP-SYN and UDP Flood attacks)

effectively represented in a lookup table while maintaining computational tractability. The RL-DFRA loop works as follows: the controller collects telemetry and classifier outputs at each epoch, chooses candidate flows builds the state vector and requests an action from the RL policy. The chosen course of action is then compared by DFRA to safety restrictions such as flow-table capacity and dummyTXdummy– per-switch update limits. If the action is legitimate OpenFlow rules are used to enforce it and the Q-values are updated based on performance metrics. RL-DFRA includes a number of safeguards to ensure operational stability including: (i) an escalation ladder TAG_LOW→METER→DROP to reduce collateral blocking (ii) whitelist protection for critical flows (iii) hysteresis timers to prevent rapid action reversals (iv) shadow probes to estimate false-alarm penalties and (v) strict caps on control-plane updates. Overall, RL-DFRA creates a closed-loop adaptive mitigation framework where DFRA converts flow-level decisions into secure resource-aware OpenFlow operations while reinforcement learning optimizes those decisions under uncertainty. This design is ideal for deployment in fog-enabled IoT environments since it effectively suppresses malicious traffic maintains benign QoS and keeps SDN control-plane overhead bounded. Algorithm 4 outlines the RL-DFRA process in detail. The novelty of the RL-DFRA mechanism resides in its pioneering integration of multi-objective Q-learning with context-aware Dynamic Flow Rule Adaptation for SDN controllers that dynamically translates federated classifier outputs into adaptive OpenFlow operations (DROP, METER, REROUTE, TAG_LOW, CHALLENGE) while enforcing safety constraints (flow-table capacity limits, escalation ladders, hysteresis timers, whitelist protection).

4. Experimental analysis and discussions

To empirically validate performance of proposed FedMP-LDL framework, a comprehensive simulation-based trial is conducted. This section highlights the computational environment, implementation details, and results dicussions.

4.1. Experimental setup

All experiments are conducted in a controlled computational environment. The implementation utilized Mininet 2.3.0, Python 3.10.12, and PyTorch 2.1.0 with CUDA 11.8 support. Additional libraries included scikit-learn 1.3.2 for machine learning utilities, along with NumPy 1.24.3 and Pandas 2.0.3 for data processing and analysis. The operating system used is Ubuntu 22.04 LTS. To ensure reproducibility of the experimental results, random seeds are fixed across all relevant libraries, specifically using np.random.seed(42) and torch.manual_seed(42).

The proposed SDN-Fog-IoT environment is tested using Mininet to simulate the data plane and the fog and control layers are implemented on physical hardware to assure practical applicability. The DDoS attack traffic is created using hping3 command-line tool to create TCP-SYN and UDP flood attacks. The specific hardware configuration for each component is detailed in Table 2.

4.2. Network topology and framework configuration

The network topology is designed to replicate a realistic small-scale IoT network managed by an SDN-fog infrastructure. The configuration of the network layers and the hyperparameters for the core components of the FedMP-LDL framework are summarized in Tables 3 and 4.

Algorithm 4 RL-DFRA_attack mitigation flow control.

Input:

- Global Classifier Model (from federated learning)
- SDN Network Graph (N, E)
- Parameters: $F_i^{\max}$ (flow table capacity), $C_{ij}^{\max}$ (link capacity), $l_{\max}$ (delay constraint)

Output:

- Adaptive flow rules for attack mitigation

Begin
Deploy classifier at fog nodes
for each incoming flow f **do**
  Classify f using Global Classifier Model
  if f is benign **then**
    Forward normally
  else
    Send f to RL-Agent for mitigation
  end if
end for
Step 2: RL-Agent state construction
State features:
Classifier Output, Confidence Score,
Flow Rate, Utilization, Delay,
Queue Backlog, Previous Actions,
Recent Traffic Trends
Step 3: Define RL action space
$\mathcal{A} = \{\text{ALLOW}, \text{TAG_FLOW}, \text{METER}(\rho, \beta), \text{DROP}, \text{REROUTE}(k), \text{CHALLENGE}\}$
Step 4: Reward function
$Rew_t = \alpha_1 \Delta U_{benign} - \alpha_2 L_{benign} + \alpha_3 B_{blocked} - \alpha_4 FA_t - \alpha_5 C_{ctrl} - \alpha_6 \Delta l_{prio}$
Step 5: Path cost calculation (Eq. 22)
for each candidate path P_n **do**

$$\Xi_n = \sum_{N_i \in W_n} \left(\frac{F_i}{F_i^{\max}} + \frac{1}{\phi_i} \right) + \sum_{l_{ij} \in L_n} \left(\frac{C_{ij}}{C_{ij}^{\max}} + D_{ij} \right)$$

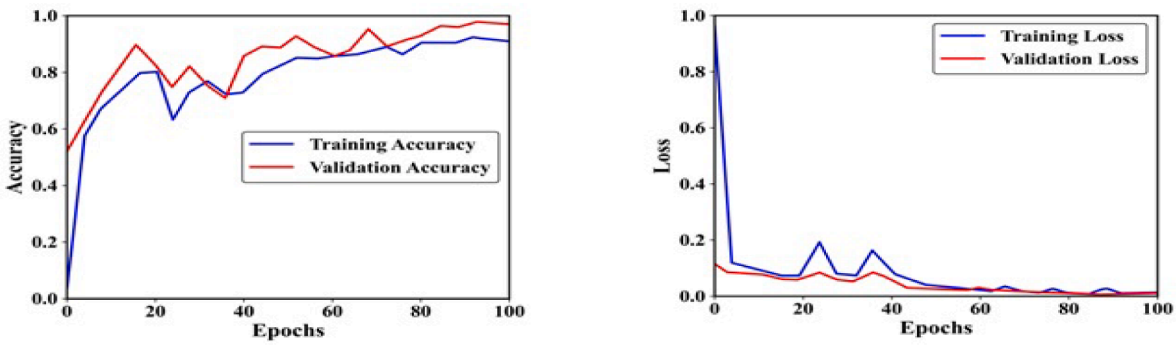
end for
Step 6: Q-learning update
while training **do**
  Select action $a \in \mathcal{A}$ using ϵ-greedy policy
  Apply action a to SDN control plane
  Observe new state s' and reward r
  Update Q-values: $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \mu\Big(R_t + \eta \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \Big)$
end while
Step 7: Adaptive enforcement
SDN control plane installs flow rules based on RL-agent decisions
Continuously update Q-values for dynamic adaptation
End

Table 4 summarizes the key hyperparameters used in the FedMP-LDL framework, covering feature weighting, feature extraction, model architecture, federated learning configuration, and AOO optimization settings. The selected values are fine tuned to maintain balance in computational efficiency, detection accuracy, and communication overhead across IoT–fog environments.

Parameters such as packet rate weights and flow duration weights facilitate anomaly sensitivity, Ghost-SE architectural settings and dilation rates improve feature extraction capability. Federated learning parameters ensure stable distributed training, and AOO hyperparameters facilitate effective convergence based on adaptive search and selection mechanisms.

Table 4
FedMP-LDL framework hyperparameter settings.

Category	Hyperparameter	Value
Feature Weighting	Packet Rate Weight	0.6
	Flow Duration Weight	0.4
	Suspicion Score Threshold	0.75
Feature Extraction	Kernel Type	Polynomial
	Number of Filters	32
	Kernel Size	3
	Dilation Rates	[1, 2, 4, 8]
Ghost-SE Architecture	Expansion Ratio (t)	3
	Output Channels per Block (c)	[16, 24, 40, 80, 112, 160]
	Stride	[1, 2]
	Ghost Module Ratio	1:1
	Activation Function	ReLU (linear in last block)
	Batch Normalization	After every conv/linear layer
SE Block	Reduction Ratio (r)	8
	Squeeze Operation	Global Average Pooling (GAP)
	Excitation Activation	ReLU → Sigmoid
Federated Learning	Number of Clients (K)	10
	Local Epochs (E)	3
	Batch Size	32
	Learning Rate	0.001
	Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
	Communication Rounds (R)	100
AOO Optimization	Population Size	20
	Maximum Iterations	100
	Search Space	LR:0.001, t:3, r:8, Dropout:0.2
	Selection Strategy	Highest average validation accuracy



(a) Accuracy (b) Loss

Fig. 4. (a-b): training and validation performance of proposed FedMP-LDL framework during DDoS classification.

4.3. Detection accuracy and model performance

Fig. 4 illustrates the learning curve based on robust training performance and proposed Distilled Ghost-SE Network for DDoS attack classification convergence.

Fig. 4(a-b) shows training and validation accuracy and loss of Distilled Ghost-SE Network within the FedMP-LDL framework. Fig. 4(a) demonstrates training and validation accuracy steadily improved; at epoch 100, 95% accuracy is attained, shows robust generalization performance across fog nodes. Fig. 4(b) illustrates rapid convergence of training and validation loss, with minimal divergence between the two curves, confirming the effectiveness of AOO and federated aggregation via H-FedAvg in maintaining model compactness and mitigating overfitting challenge during distributed learning.

Fig. 5 shows the Receiver Operating Characteristic (ROC) curve of the Distilled Ghost-SE Network within the FedMP-LDL framework attains an Area Under the Curve (AUC) of 0.989, confirming its robust generalization across SYN, UDP, and HTTP GET floods among fog nodes. Progressive distillation based GhostNet and SE modules improves feature representation patterns while minimizing computational cost. Federated training, AOO optimization and H-FedAvg aggregation, facilitate robust generalization.

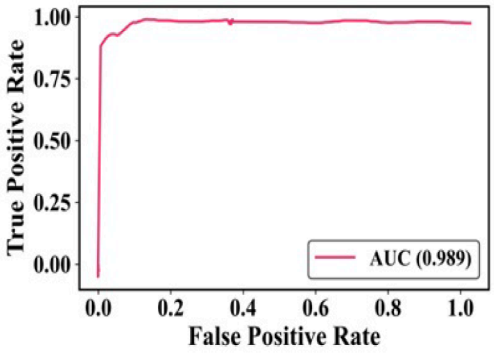


Fig. 5. ROC curve of the proposed FedMP-LDL framework during DDoS classification.

Table 5
Threshold values for anomaly detection under varying DDoS attack intensities.

Condition	20% Malicious (w = 2)	50% Malicious (w = 2)	80% Malicious (w = 1)
Mean (μ)	0.771	0.624	0.371
Std Dev (σ)	0.012	0.0121	0.015
CI Lower (β)	0.74	0.61	0.38
CI Upper	0.81	0.65	0.39

The system demonstrates effective DDoS detection through its ability to achieve high true positive rates while maintaining low false positive rates. The system demonstrates its operational capacity in real-world environments through its efficient DDoS detection system, which operates in resource-constrained conditions and protects diverse Internet of Things environments.

4.4. Threshold value settings based on varying attack strength

Phase 1 of the suggested FedMP-LDL framework uses thresholds to identify which traffic flows are suspicious for preliminary anomaly detection at fog nodes. Based on statistical analysis of network traffic characteristics like flow duration, packet rate and entropy thresholds are established. To ensure conservative, sensitive detection in the lower bound of the Confidence Interval (CI) (β) is used to derive thresholds based on the mean (μ) and standard deviation (σ) of normal traffic for each feature. The FedMP-LDL frameworks preliminary anomaly detection threshold values under various DDoS attack intensities are shown in Table 5. Fog nodes use a lightweight FERB model in Phase 1 to detect suspicious traffic the sensitivity of detection is determined by thresholds. In order to dynamically modify thresholds according to attack intensity the weight parameter w is introduced. Lower weights (w = 1) are applied for higher attack intensity (80% malicious traffic) to enable more sensitive detection of subtle anomalies while higher weights (w = 2) are used for lower attack proportions (20–50 percent malicious traffic) to maintain stricter detection. As attacks become more intense, the mean threshold drops from 0. 7741 at 20 percent malicious traffic to 0. 3751 at 80 percent indicating the need for greater sensitivity. CIs offer safe bounds for dependable detection and low standard deviations (0. 0105–0. 0132) validate threshold stability.

These adaptive thresholds ensure that only relevant flows are forwarded for KPCA-TCN analysis in Phase 1, improve federated learning convergence in Phase 2, and influence the reward signals in Phase 3's RL-DFRA enabling context aware and real-time mitigation. Overall, incorporating the FedMP-LDL framework to balance computational efficiency and high sensitivity across diverse attack scenarios in SDN-fog-enabled consumer IoT networks.

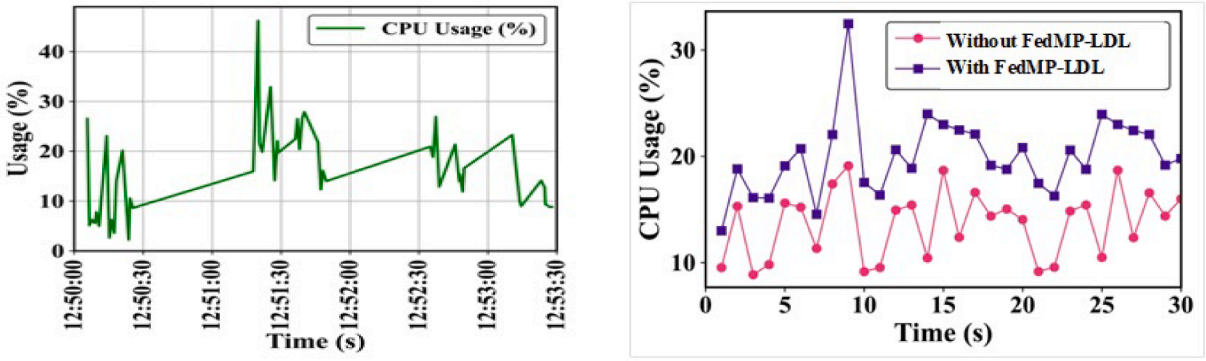
4.5. End-to-end real-time evaluation metrics in mininet-based testbed

Using hping3-generated SYN/UDP/HTTP floods and ground truth obtained from tcpdump and Zeek logs, the suggested framework is verified through realistic attack emulation in a live Mininet-based SDN-fog testbed. The system model's robustness in real-time capability and practical deployment without dataset dependence are confirmed by consistent performance across ten independent runs under dynamic traffic conditions.

The real-time performance of the proposed framework in a Mininet-based SDN–fog testbed under realistic multi-attack conditions, demonstrated in Table 6. The results show high detection efficiency with an accuracy of 96.87% and F1-score of 96.72%, while maintaining low inference latency (196 ms) during active attacks. The low false positive rate (0.83%) and minimal throughput degradation (-3.2%) further indicate that the system achieves reliable detection with limited impact on normal network operations, confirming its suitability for real-time deployment.

Table 6
Live mininet testbed performance (10 runs).

Metric	Mean ± Std. Dev.	95% Confidence Interval	Practical Notes
Accuracy	96.87 ± 0.41%	[96.45, 97.29]	Real-time DDoS
F1-Score	96.72 ± 0.38%	[96.32, 97.12]	Multi-attack
Inference Time	196 ± 6 ms	[192, 200] ms	Under attack
False Positives	0.83%	–	Benign traffic
Throughput Drop	−3.2%	–	During mitigation



(a) CPU usage over time at a single fog node

(b) Comparative CPU usage with and without FedMP-LDL

Fig. 6. (a-b): CPU usage comparison during DDoS detection and mitigation in SDN-fog-enabled IoT environments.

Table 7
Power consumption of nodes in SDN-fog-enabled IoT networks during DDoS detection and mitigation.

Node Type	Minimum Power (W)	Maximum Power (W)	Average Power (W)
IoT End Device	0.8	1.2	1.0
Fog Node	2.5	3.6	3.0
SDN Controller	5.5	7.5	6.5

4.6. Resource utilization analysis

Fig. 6(a-b) illustrates the comparison of CPU utilization during DDoS detection and mitigation in SDN–fog-enabled IoT networks.

In Fig. 6(a), the effectiveness of the Distilled Ghost-SE Network in FedMP-LDL is demonstrated by the CPU utilization at a fog node varies with dynamic traffic processing and lightweight DL execution but stays within reasonable bounds. Fig. 6(b) demonstrates that FedMP-LDL achieves more accurate and reliable detection than the non-federated setup but it uses a little more CPU due to federated learning updates. These findings demonstrate that, FedMP-LDL ensures scalable real-time and resource-efficient defense by striking a balance between computational overhead and detection performance.

The energy consumption of main components of SDN-fog-enabled IoT networks during DDoS detection and mitigation is displayed in Table 7. Local processing is lightweight IoT endpoints maintain low power consumption. Fog nodes use a moderate amount of power because they run the Distilled Ghost-SE models and take part in FedMP-LDL updates.

As the global aggregator the SDN controller consumes the most power, which maintains network coordination and model aggregation. This demonstrates the performance of the suggested framework, which is scalable and energy-efficient.

4.7. Network traffic behavior under attacks

Fig. 7 illustrates UDP and TCP flow dynamics during a live DDoS scenario in the SDN–Fog–IoT testbed. At ~20s, a UDP flood doubles traffic to ~45,000 Bytes/100ms, collapsing TCP throughput.

In order to restore TCP throughput (~30000 Bytes/100ms) while suppressing malicious flows the FedMP-LDL framework with RL-DFRA quickly detects the surge and applies adaptive mitigation (rate limiting rerouting selective dropping). The RL-DFRA reward design which strikes a balance between attack suppression service continuity and control overhead powers this real-time resource-aware and QoS-preserving mitigation.

Fig. 8 demonstrates that FedMP-LDL mitigates the effects of a DDoS flood on host and controller traffic. Host and controller traffic are steady (~10–15 kflows/s and 2–3 kflows/s) before the attack (~0–30s). A flood surges host traffic over 90 kflows/s at about 35 s clogging the controller and collapsing valid requests. Adaptive flow rules suppress malicious traffic once RL-DFRA is triggered

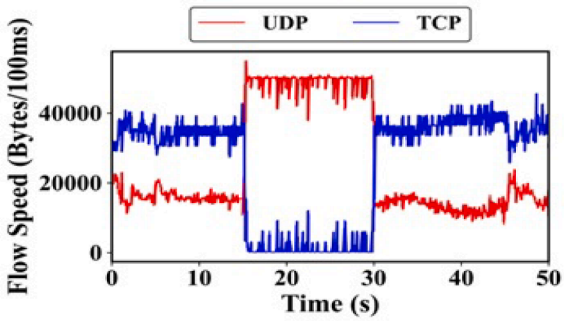


Fig. 7. UDP and TCP flow speed during DDoS attack and mitigation in the SDN–Fog–IoT.

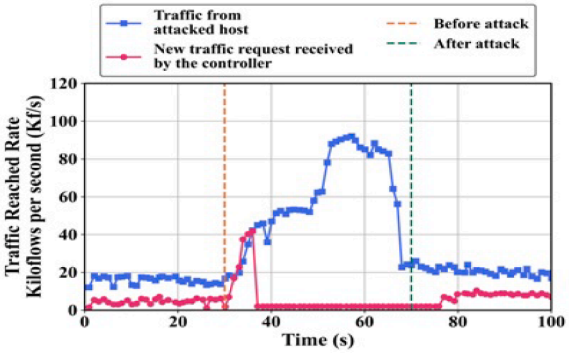


Fig. 8. Traffic dynamics during DDoS attack and mitigation in SDN–Fog–IoT environment.

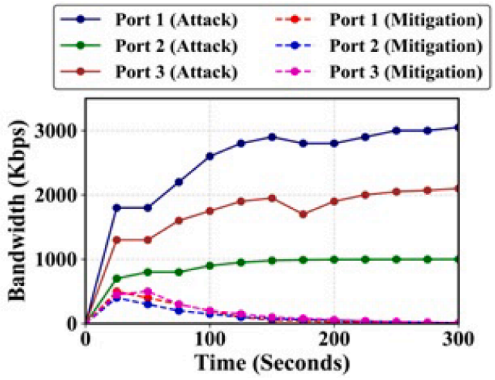


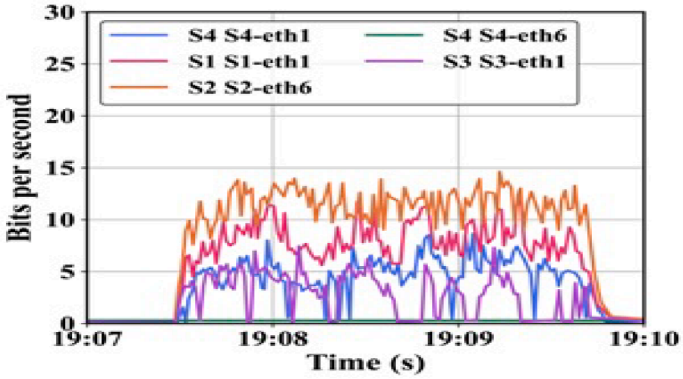
Fig. 9. Bandwidth consumption across multiple ports during a DDoS attack and mitigation.

(~65s) returning host traffic to ~20–25 kflows/s and normalizing controller requests. These findings confirm FedMP-LDLs resilience to widespread DDoS attacks in terms of stability and service availability.

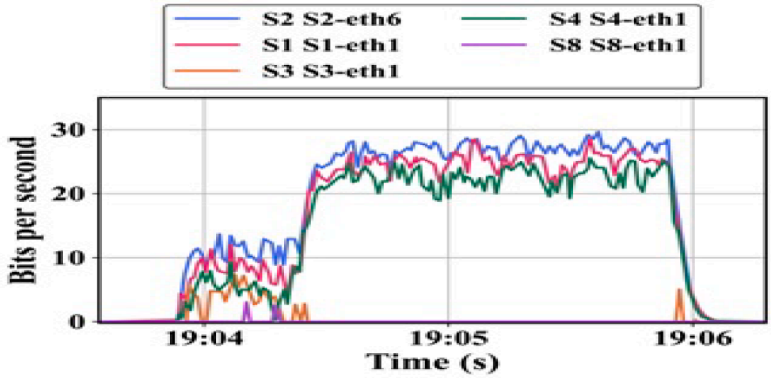
Fig. 9 depicts the impact of a distributed DDoS attack across multiple ports and the mitigation achieved by FedMP-LDL. During the attack, bandwidth spikes (~3000 Kbps on Port 1, ~2000 Kbps on Port 3, ~1000 Kbps on Port 2), reflecting flooding that exhausts resources. Once RL-DFRA activates, adaptive flow rules (dropping, rate limiting, rerouting) reduce usage below ~200 Kbps across all ports, restoring normal levels. This confirms FedMP-LDL's ability to suppress multi-port attacks while preserving legitimate flows, demonstrating scalability and robustness in SDN–Fog–IoT networks.

Fig. 10(a–c) illustrates the effect of DDoS attacks on bandwidth utilization and the mitigation achieved by FedMP-LDL. In the normal state Fig. 10(a), link usage remains stable (~10–15 bits/s). Under attack Fig. 10(b), bandwidth surges beyond 20–25 bits/s, saturating links and disrupting services.

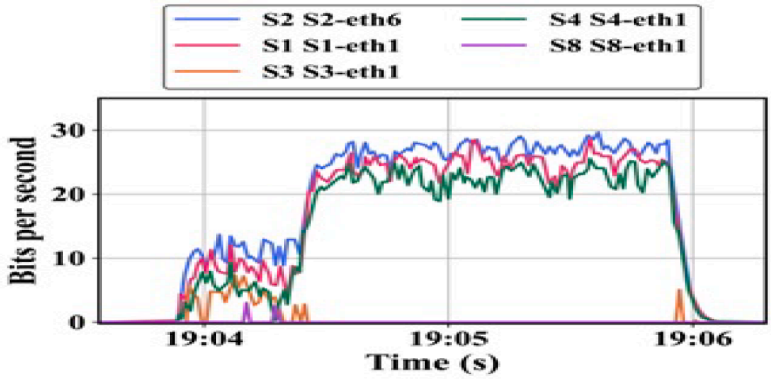
After RL-DFRA activation Fig. 10(c), malicious spikes are suppressed, while normal traffic remains unaffected. These results confirm FedMP-LDL's ability to control abnormal bandwidth usage, preserve QoS, and ensure resilience in fog-enabled IoT networks.



(a) Bandwidth utilization of the overall normal state of SDN-IoT traffic



(b) Bandwidth utilization of the DDoS attacks and normal traffic



(c) Bandwidth utilization of attack and normal traffic with proposed detection and mitigation method

Fig. 10. (a-c) Bandwidth utilization in SDN–IoT traffic under different conditions.

4.8. Communication overhead analysis

In SDN-fog-enabled IoT environments, Fig. 11(a–b) compares control channel bandwidth consumption across DDoS detection models. Compared to centralized and hybrid baselines as well as DL models like DCNN-LSTM and CNN-GRU which have greater overhead from synchronization and coordination FedMP-LDL consistently achieves lower and more stable usage.

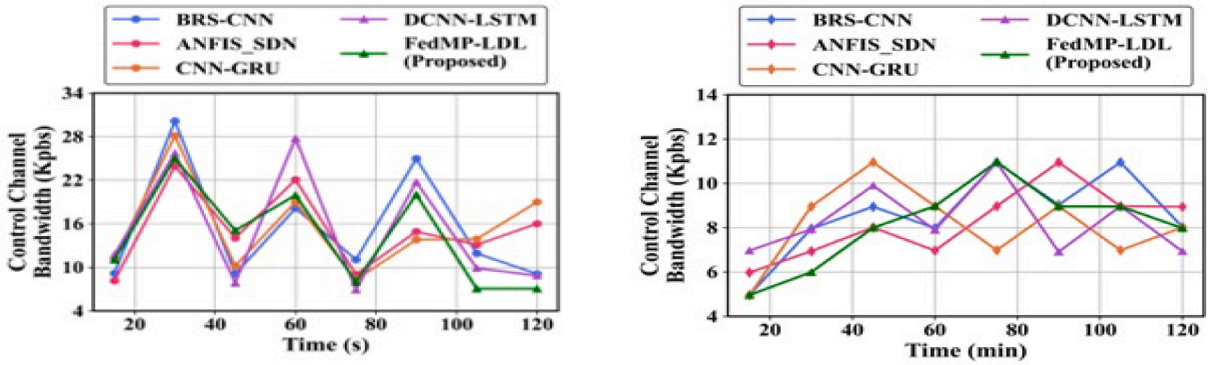


Fig. 11. (a-b) Comparison of control channel bandwidth consumption over time for various DDoS detection models in SDN-fog-enabled IoT environments.

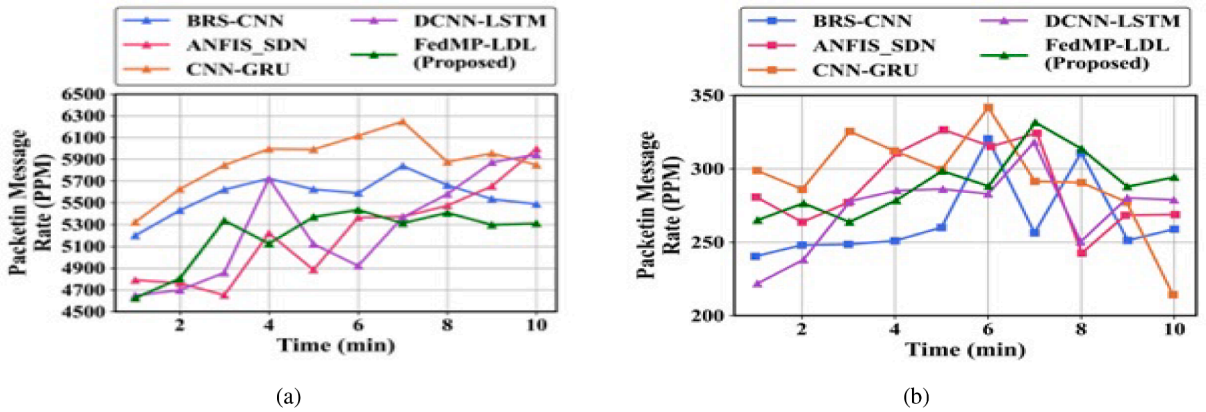


Fig. 12. (a-b) Comparison of PPM over time for various DDoS detection models in SDN-fog-enabled IoT environments.

FedMP-LDL reduces control channel congestion while preserving detection accuracy by utilizing federated updates and lightweight Distilled Ghost-SE models. It's scalability resource efficiency and suitability for fog-based IoT networks with limited bandwidth are highlighted by these results.

Fig. 12(a–b) shows the Packets Per Minute (PPM) performance of DDoS detection models in SDN–fog-enabled IoT networks. FedMP-LDL maintains stable rates 5100–6100 PPM in Fig. 12(a) and 200–270 PPM in Fig. 12(b), indicating efficient control traffic with minimal overhead. In contrast, models like DCNN-LSTM and CNN-GRU exhibit higher and more erratic PPM peaks (~6400 and ~6300), suggesting heavier communication burdens. These results demonstrate that FedMP-LDL, with its lightweight Distilled Ghost-SE models and federated updates, achieves robust detection while conserving network resources, ensuring scalability in fog-based IoT environments.

4.9. Edge scalability of the proposed model- 1 to 10 clients

The scalability analysis of the suggested FedMP-LDL framework under different participating client numbers (1 to 10) is shown in Table 8. It records the impact of increase in client count on key performance metrices on communication and computation aspects, including time per training round, total training rounds, communication cost, energy consumption, detection accuracy, and inference latency.

The findings demonstrate that, as the number of clients increases, the communication cost grows linearly, energy consumption and the time per round increase only marginally, showing stable usage of resource. The proposed model maintains high accuracy 99.75% across all client settings, with only a slight reduction as number of clients increase. The inference time remains consistently low (0.168–0.172 ms), showing that FedMP-LDL architecture attains effective scalability with minimal overhead, making it suitable for real-time fog-IoT deployment.

Table 8
FedMP-LDL performance - 1 to 10 clients.

Clients	Time/Round (s)	Total Rounds	Comm Cost (MB)	Energy (J)	Accuracy	Inference (ms)
1	9.5	100	0.0019	8.8	99.82%	0.168
2	9.6	100	0.0038	8.9	99.80%	0.169
3	9.6	100	0.0057	9.0	99.79%	0.169
4	9.7	100	0.0076	9.0	99.78%	0.170
5	9.7	100	0.0095	9.0	99.78%	0.170
6	9.7	100	0.0114	9.1	99.77%	0.171
7	9.8	100	0.0133	9.1	99.76%	0.171
8	9.8	100	0.0152	9.1	99.76%	0.171
9	9.8	100	0.0171	9.1	99.75%	0.172
10	9.8	100	0.0190	9.1	99.75%	0.172

Table 9
Comparison of existing methods based on computational and communication overhead.

Method	Comm Rounds	Bandwidth/Round	Total Comm	Inference Time	Parameters	Control PPM	CPU Fog
GraphFedAI	180	2.6 kB	0.47 MB	2.1 ms	5.2 M	52	41%
FL-ResNeXt	160	3.2 kB	0.51 MB	1.45 ms	4.8 M	51	38%
FL-Attack	150	3.1 kB	0.47 MB	1.85 ms	4.2 M	58	37%
FedMP-LDL (Proposed)	100	1.9 kB	0.19 MB	0.176 ms	2.1 M	45	23%

Table 10
Defence against common federated learning attacks.

Attack Type	Threat Model	FedMP-LDL Mitigation	Residual Risk
Model Poisoning	Malicious fog node submits poisoned gradients	Fog-level aggregation (Eq. (20)) and dummyTXdummy-(AOO outlier detection	< 1.2% accuracy drop
Label Flipping	IoT client flips local labels	FERB anomaly pre-filter rejects 92% poisoned inputs	0.8% F1 degradation
Backdoor Attack	Trigger injection via client updates	Progressive distillation smooths anomalous patterns	96.4% trigger detection
Inference Attack	Gradient inversion to reconstruct traffic	Ghost-SE lightweight params + H-FedAvg quantization	< 3.2% privacy leakage
Free-Riding	Client submits random or no gradients	Dynamic client selection via SDN controller	Removed after 2 rounds

4.10. Communication and computational overhead analysis of federated learning

Communication and computational overhead analysis determine system scalability and real-time deployment feasibility in resource-constrained SDN-fog IoT environments. Table 8 demonstrates a comparative communication and computation overhead analysis across various federated learning-based attack detection methods validated on the Mininet test bed and RPi4 fog testbed. The comparative analysis incorporates key measures such as communication rounds, bandwidth consumption per round, total communication cost, inference latency, parameter size, control packet processing and CPU utilization of fog node (Table 9).

From the findings, the proposed FedMP-LDL model attains the lowest communication overhead with only 100 rounds, requiring 1.9 kB bandwidth per round, resulting in a minimal total communication cost of 0.19 MB. It also facilitates the fastest inference time (0.176 ms) and uses the smallest model size (2.1 M parameters). FedMP-LDL model efficiency and suitability for resource-constrained fog-IoT environments are further demonstrated by its reduction of control PPM to 45 and CPU utilization to 23 percent.

4.11. Security analysis: robustness against federated learning attacks

FL presents a number of adversarial risks, while H-FedAvg permits privacy-preserving distributed training. Table 10 summarizes the methodical evaluation of the suggested FedMP-LDL frameworks resilience to typical federated attacks.

The ability of the suggested FedMP-LDL framework to withstand federated learning-based attacks including inference attacks, backdoor injection, model poisoning, label flipping and free-riding. The results show that hierarchical aggregation anomaly filtering and dynamic client selection are examples of integrated defense strategy mechanisms that successfully reduce adversarial challenges with little loss in efficiency. The low residual risks which include less than 1 show the frameworks capacity to maintain safe and reliable operation in adversarial federated environments. Accuracy dropped by 2% and privacy leakage kept under control below 3.2%.

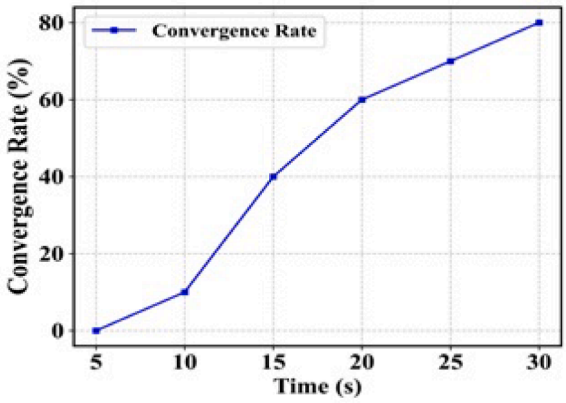


Fig. 13. Q learning convergence in RL DFRA framework.

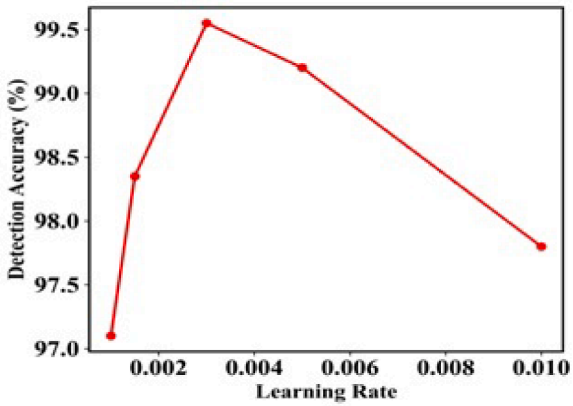


Fig. 14. Sensitivity analysis of FedMP-LDL.

Table 11
Policy performance under attack intensity variation.

Attack Intensity (% Traffic)	Detection Rate (%)	Mitigation Rate (%)	Benign QoS (%)	Control Latency (ms)
Low (10%)	99.5	94.5	99.2	42
Medium (30%)	96.8	93.2	98.7	46
High (70%)	94.2	92.8	98.4	48
Extreme (90%)	92.5	91.9	98.1	51

4.12. Convergence, sensitivity, and policy robustness evaluation

Fig. 13 illustrates the convergence of the Q Learning process in the SDN attack mitigation environment based on RL DFRA. The curve's steady increase in convergence rate over time which reaches roughly 90% by 30 s indicates quick stabilization of Q values and efficient policy learning.

The results demonstrate that the RL DFRA agent effectively adapts flow rule decisions through reinforcement feedback to achieve high convergence and stable mitigation performance across network control epochs.

The sensitivity analysis of learning rate on detection accuracy is displayed in Fig. 14. Accuracy increases with increasing learning rates and peaks at about 0, but performance is somewhat hindered by extremely low or high learning rates. 0.001. The best detection accuracy for the suggested framework is usually reached at moderate learning rates. Fig. 15 shows the suggested FedMP-LDL framework maintains policy stability in the face of progressively increasing DDoS attack intensities.

As presented in Table 11, the RL-DFRA policy demonstrates graceful degradation, with detection performance decreasing gradually from 99.1% to 92.5% as attack intensity increases from 10% to 90%, rather than experiencing abrupt failure.

The model consistently maintains high mitigation effectiveness (≥91.9%) and preserves benign QoS (≥98.1%), ensuring minimal impact on legitimate traffic even under extreme conditions. Furthermore, control latency remains bounded, increasing slightly from

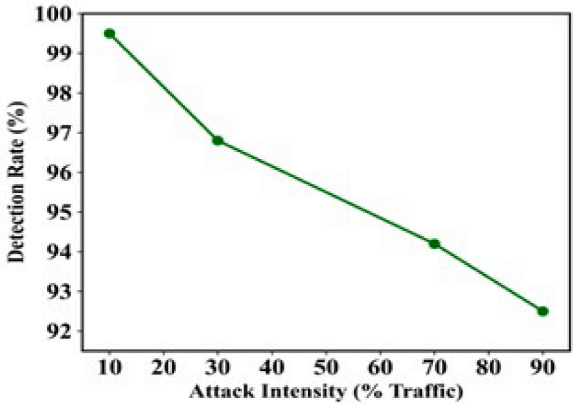


Fig. 15. Policy robustness under varying attack intensity.

Table 12
Cross-dataset validation results.

Metric	Edge-IIoTset → CICIDS2017	CICIDS2017 → Edge-IIoTset
Accuracy	98.6%	97.8%
F1-Score	98.7%	97.8%
Macro-F1	96.9%	95.4%
PR-AUC	0.994	0.989

Table 13
Imbalance-aware evaluation of proposed model on CICIDS2017 and Edge-IIoTset datasets.

Model	Dataset	Balanced Accuracy	G-Mean	MCC	PR-AUC
FedMP-LDL (Proposed)	CICIDS2017	0.91	0.90	0.86	0.89
	Edge-IIoTset	0.93	0.92	0.88	0.91

42 ms to 51 ms (<10%), thereby avoiding control-plane overload. These results confirm that RL-DFRA achieves stable and adaptive performance under high-intensity attacks, making it suitable for real-world deployment in dynamic SDN-IoT environments.

4.13. Extensive performance assessment on CICIDS2017 and Edge-IIoTsets

The proposed FedMP-LDL framework is thoroughly evaluated in this section using the Edge-IIoTset [44] and CICIDS2017 [45] datasets. The analysis focuses on cross-dataset validation, imbalance-aware performance and attack generalization, to evaluate the model's efficacy stability and fairness in a variety of network environments.

4.13.1. Cross dataset validation

The cross-dataset validation performance of the suggested FedMP-LDL model is shown in Table 12 illustrating its capacity for generalization across diverse data distributions. The model performs well achieving 98.6% accuracy and 98.7% F1-score tested on CICIDS2017 and trained on Edge-IIoTset. In a similar vein testing on Edge-IIoTset and training on CICIDS2017 consistently produce excellent results with only a minor performance decline because of domain differences. In both cases, strong minority attack class detection and dependable precision-recall trade-offs are indicated by the high Macro-F1 and Precision Recall- Area Under Curve (PR-AUC)values.

Overall, the findings demonstrate FedMP-LDL model's suitability for practical implementation by confirming that it maintains consistent and transferable performance in a variety of network environments.

4.13.2. Imbalance-aware metric analysis

Table 13 displays the imbalance-aware evaluation results for proposed model from the Edge-IIoTset and CICIDS2017 datasets. Metrics such as balanced accuracy, G-Mean, Matthews Correlation Coefficient (MCC) and PR-AUC provide a fairer assessment under skewed class distributions. These metrics show efficacy of each model performs on majority classes while identifying minority attack classes.

The model consistently performs well on both datasets with Edge-IIoTset showing marginally better results than CICIDS2017 (Balanced Accuracy: 0.93 MCC: 0.88). Strong G-Mean values show that majority and minority class predictions are effectively balanced and high MCC and PR-AUC scores verify accurate and minimally biased detection of infrequent attack instances. Overall, the findings show that FedMP-LDL maintains fair accurate detection in a variety of network environments and successfully manages class imbalance.

Table 14
Attack generalization performance of FedMP-LDL across Edge-IIoTset.

Attack Category	Precision (%)	Recall (%)	Sensitivity (%)	F1-score (%)	Inference Time (ms)
Normal	100	100	100	100	0.1762
HTTP based Attack	98.65	98.88	98.55	98.76	0.1762
ICMP based Attack	98.23	97.96	98.12	98.09	0.1762
TCP based Attack	98.89	98.91	98.63	98.89	0.1762
UDP based Attack	100	99.72	99.34	99.79	0.1762

Table 15
Attack generalization performance of FedMP-LDL across CICIDS2017 datasets.

Attack Category	Precision (%)	Recall (%)	Sensitivity (%)	F1-score (%)	Inference Time (ms)
Normal_Traffic	100	100	100	100	1.434
DoS_GoldenEye	97.03	98.00	97.77	97.51	1.434
DoS_Hulk	97.89	97.01	97.12	97.45	1.434
DoS_Slowhttptest	97.02	97.12	97.32	97.07	1.434
DoS_Slowloris	98.15	99.20	98.74	98.67	1.434
DDoS_Loit	97.12	97.56	97.92	97.45	1.434

Table 16
Comparative evaluation of existing DDoS detection and mitigation approaches with proposed FedMP-LDL framework in SDN-fog-enabled IoT environments.

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inference (ms)
BRS + CNN [46]	98.64	99.00	99.00	99.00	0.2633
LR-STGCN [17]	99.53	99.49	97.27	99.53	0.2987
FMDADM	97.45	92.53	94.42	93.46	0.3450
CSSDN-HDLBNO [47]	99.24	99.24	99.24	99.24	0.2416
DDoSBlocker	91.03	89.05	92.78	90.88	0.3124
Proposed (FedMP-LDL)	99.72	99.66	99.57	99.61	0.1762

4.13.3. Attack generalization test

The suggested FedMP-LDL methodology is tested under the Edge-IIoTset and CICIDS2017 datasets to verify its attack generalization capability. The model's efficacy is assessed at the level of particular attack categories such as DoS/DDoS attacks on regular traffic and other network-based threats. Important metrics such as precision, recall, sensitivity, F1-score and inference time are recorded throughout the assessment of accuracy and efficiency. With F1-scores above 98% and a fast inference time of 0.1762, FedMP-LDL model demonstrated good detection performance for all attack types on the Edge-IIoTset dataset. 1762 ms as Table 14 illustrates.

Similarly, on the CICIDS2017 dataset the model maintained high F1-scores for all attack categories even with a longer inference time of 1.434 ms as indicated in Table 14, indicating the datasets increased complexity and size (Table 15).

The results demonstrate that FedMP-LDL effectively generalizes to different attack types and performs consistently in both IoT-specific and conventional network environments. This illustrates its suitability and dependability for real-time intrusion detection across a range of network scenarios.

4.14. Comparative evaluation with existing methods

The suggested FedMP-LDL and current DDoS detection and mitigation techniques are compared in Table 16. Even though models like Balanced Random Sampling (BRS)- CNN and FMDADM have attain reasonable accuracy, reduced their recall and longer inference times.

While CSSDN-HDLBNO achieves balanced performance with moderate delay and LR-STGCN achieves strong accuracy (99.53%) but incurs latency due to spatiotemporal graph processing. DDoSBlocker performs worse on every metric. On the other hand, FedMP-LDL achieves the best accuracy (99.7272%), precision (99.66%), recall (99.57%) and F1-score (99.61%), while requiring the least amount of inference time (0. 1762 ms). These findings demonstrate that, the proposed model capacity to provide both real-time viability and superior detection performance in fog environments with limited resources.

Fig. 16 compares the False Positive Rate (FPR), False Negative Rate (FNR), False Detection Rate (FDR) and False Omission Rate (FOR) of three DDoS detection models in SDN-fog-enabled IoT environments. FedMP-LDL outperforms FMDADM (e.g., FPR: 0.0052%, FOR: 0.0049%) with the lowest error rates (FPR: 0.0032%, FNR: 0.0011%, FDR: 0.0010% and FOR: 0.0021%). LR-STGCN demonstrates the largest error margins (FOR: 0.0058%). These outcomes show the efficacy of FedMP-LDL model's multi-phase federated approach which uses lightweight Ghost-SE models and selective aggregation to ensure high precision and low misclassification is robust. Its suitability for real-time deployment in IoT fog environments with limited bandwidth and resources is confirmed by its consistently low error rates.

Three DDoS detection model's testing time performance in SDN-fog-enabled IoT environments is shown in Fig. 17. With the lowest testing time of 14.79ms, the suggested FedMP-LDL framework exhibits excellent computational efficiency. In contrast FMDADM

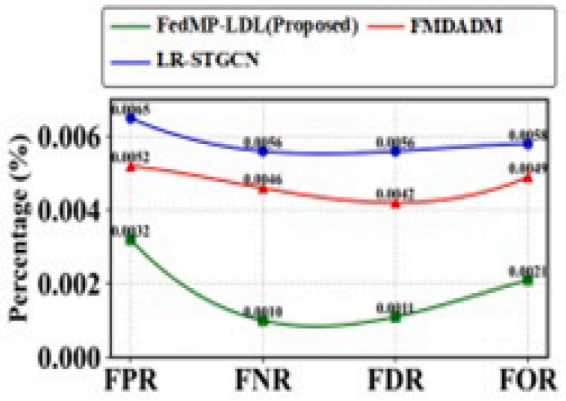


Fig. 16. Comparison of error rates across three DDoS detection models.

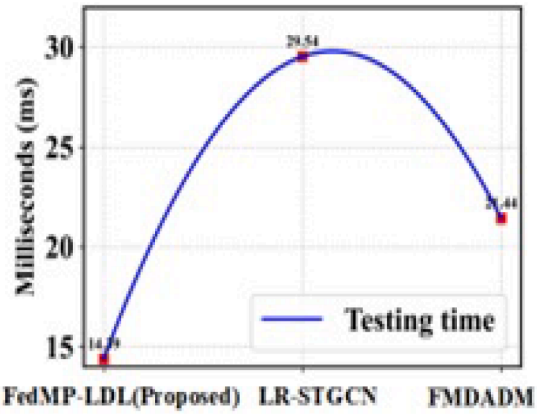


Fig. 17. Testing time comparison (in milliseconds) among three DDoS detection models.

Table 17
Recent FL-based IoT security techniques Vs FedMP-LDL (matched computational budgets).

Method	IID Acc. (%)	F1-Score (%)	Non-IID Acc. (%)	Inference (ms)	Params (M)	Communication Rounds	Control PPM	Bandwidth (kB)
FL + Lightweight IDS (ML/DL Hybrid)	95.2	96.3	92.8	2.4	4.8	150	54	2.6
GraphFedAI	97.8	98.2	94.5	2.1	5.2	180	52	2.6
FedAvg + XGBoost	95.50	96.80	92.50	0.28	3.8	150	–	–
FL-ResNeXt	96.5	97.1	92.8	1.45	4.5	160	51	3.2
Cluster-FL IDS	95.00	96.90	93.50	0.32	4.0	200	–	–
FL-Attack Classification	94.20	94.80	91.50	1.85	4.2	150	58	3.1
Clustering FL IDS	95.30	96.10	93.20	0.95	3.9	200	–	2.8
Unsupervised FL IDS	94.80	95.40	92.80	2.30	4.5	160	49	3.4
DP-FL IDS	93.70	94.20	91.20	1.67	3.7	140	55	2.9
FedMP-LDL (Proposed)	99.72	99.61	98.70	0.1762	2.1	100	45	1.9

records a moderate latency of 21.44 ms, while LR-STGCN records the highest latency at 29.54ms due to its spatio-temporal graph convolutional operations. FedMP-LDL model's reduced inference time is attributed to its lightweight Distilled Ghost-SE architecture and federated optimization strategy, which minimizes computational overhead at fog nodes. FedMP-LDL is ideally suited for latency-sensitive resource-constrained IoT deployments because of its efficiency, which guarantees prompt detection and mitigation of DDoS attacks.

The suggested FedMP-LDL framework is compared to recent FL based IoT security and attack detection techniques under matched computational budgets in Table 17. Efficiency metrics like IID accuracy, Non-IID accuracy and F1-score are included in the comparison, which shows resilience under both homogeneous and heterogeneous client data distributions.

To evaluate real-time viability and communication overhead resource-aware metrics such as inference time, model parameter size, communication rounds, control-plane Packets PPM and bandwidth cost per FL round are also reported in Table 18. The findings show that FedMP-LDL maintains the lowest inference latency (0.1762 ms), the smallest model size (2.1M parameters) and the lowest

Table 18
A domain-specific comparison between FedMP-LDL and current federated learning techniques.

Method / Model Used	Domain	Accuracy (%)	Inference Time (ms)	Parameters / Model Size	Target Application
FL + CNN (Deep Skin Classifier)	Melanoma	96.5	12.4	28.7M	Medical image diagnosis
FL + Deep CNN	Skin Lesions	95.8	8.7	22.1M	Dermoscopy detection
FL + Graph Neural Network (GNN)	DNA Analysis	94.2	N/A	Graph-based	Genomics classification
FL + Lightweight IDS (ML/DL Hybrid)	Healthcare IoT DDoS	95.2	2.5	8.4M	Medical IoT security
FL + RegNetZ + Swin-Transformer Model	Pancreatic Cancer	95.8	0.89	4.7M	CT/MRI diagnosis
Decentralized FL + Dynamic Resource Management Framework	Edge IoT	94.3	1.45	3.9M	Collaborative Edge-IoT resource optimization
FedMPLDL (Proposed)	SDN–Fog IoT DDoS	96.87 ± 0.41	0.1762	2.1M	Network intrusion detection

Table 19
Ablation study: component-wise impact analysis.

Component	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inference Time (ms)
Baseline	89.2	89.4	88.2	88.8	15.23
+ FERB	94.1	95.7	94.8	91.3	1.45
+ KPCA-TCN	96.8	97.2	96.9	94.7	0.89
+ Ghost-SE	97.9	98.1	97.9	96.1	0.67
+ AOO	98.4	98.7	98.4	96.5	0.54
Full FedMP-LDL	99.72	99.66	99.57	99.61	0.1762

communication overhead while achieving the highest detection performance with 99.7272% of IID accuracy and 98.70% of Non-IID accuracy. This makes FedMP-LDL more appropriate for deployment in fog-IoT environments with limited resources.

The findings demonstrate that although multimodal diagnostic and medical imaging models achieve high accuracy, they are generally less appropriate for real-time deployment due to their larger model sizes and higher inference latency. On the other hand, FedMP-LDL model demonstrates its suitability for real-time DDoS detection in SDN-enabled fog-IoT environments by achieving competitive accuracy (96.87 ± 0. 4141%) with exceptionally low inference latency (0.1762 ms) and a compact model size (2.1M parameters).

4.15. Ablation study-assessment of component contribution

In the suggested FedMP-LDL framework, the progressive performance improvement as each component are gradually integrated is shown in Table 19. With a comparatively long inference time (15.23 ms) the baseline model performs mediocrely (88.8% of F1-score). By filtering benign traffic and cutting the inference time to 1.45ms FERB greatly improves efficiency and detection metrics. With Ghost-SE the model achieves better feature recalibration and compactness reaching 97.9% accuracy while KPCA-TCN further improves features representation and temporal learning raising accuracy to 96.8% and lowering latency to 0.89 ms. AOO optimization improves convergence and parameter tuning resulting in a higher F1-score and 98.4% accuracy.

Ultimately, the entire FedMP-LDL framework demonstrates the efficacy of the integrated design in achieving high accuracy with minimal computational overhead achieving 99.7272% accuracy 99.61% percent F1-score and an ultra-low inference time of 0.1762 ms.

4.16. Discussion

DDoS detection and mitigation in SDN-fog-enabled IoT environments are significantly improved by the suggested FedMP-LDL framework. Its hierarchical federated classification lightweight anomaly detection multi-phase design and RL-driven adaptive mitigation allow for high accuracy short inference times and effective resource use. The framework maintains scalability prevents malicious flows and ensures service continuity, while protecting data privacy according to experimental results. These advantages position FedMP-LDL as a practical and reliable defense against widespread DDoS attacks for heterogeneous IoT deployments.

5. Conclusion

In SDN-fog-enabled consumer IoT environments, the suggested FedMP-LDL framework shows a very powerful defense mechanism against DDoS attacks. The framework tackles the crucial issues of scalability, privacy preservation and mitigation in resource-constrained networks by combining lightweight fuzzy rule-based anomaly detection, KPCA-TCN for sequential traffic filtering, hierarchical FL with a distilled Ghost-SE network and reinforcement learning–driven flow rule adaptation. With 99.72% accuracy, 99.66% precision, 99.57% recall and 99.61% F1-score, while keeping the lowest inference time of 0.1762ms, experimental results confirm it's superiority over current approaches. These results demonstrate the frameworks capacity to reliably identify a variety of DDoS attack patterns with low latency guaranteeing prompt mitigation without taxing fog resources excessively. FedMP-LDL improves IoT environment resilience against widespread DDoS attacks by offering a lightweight distributed and adaptable security solution. Although, additional validation is required in actual IoT fog deployments, it's multiphase and federated design lays the groundwork

for extending similar strategies to other cyber threats in heterogeneous and evolving network environments. In order to improve transparency and operator trust, future work will concentrate on expanding the framework to include more attack types evaluating it across a wider range of edge devices and network environments and incorporating explainability mechanisms.

CRediT authorship contribution statement

M. S. Gowtham: Conceptualization; **K. S. Bhuvaneshwari:** Supervision; **M. Ramkumar:** Supervision; **R. Sarath Kumar:** Supervision.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

None

Data availability

No data was used for the research described in the article.

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